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Eliminate-Then-Select: A human-centric reasoning framework for educational question answering with LLMs

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ABSTRACT

Large Language Models (LLMs) demonstrate strong performance in educational question answering, yet they often lack human-like reasoning and explainability—particularly in tasks requiring multi-step elimination before selection. We introduce the Eliminate-Then-Select (ETS) framework, a modular system that guides LLMs to emulate the human process of ruling out incorrect answers step-by-step before selecting the final answer. ETS operates through four coordinated modules: (A) per-option chain-of-thought reasoning with confidence estimation; (B) elimination via threshold or classifier; (C) dynamic strategy control, including self-debate and iterative refinement; and (D) answer-explanation alignment with calibrated confidence. Our framework is evaluated across six educational QA benchmarks (PIQA, RACE, ScienceQA, Open-BookQA, PubMedQA and EduExamQA), covering both general knowledge and domain-specific tasks. ETS achieves an average accuracy improvement of 4.3% over strong LLM baselines, with significantly better explanation quality measured by ROUGE-L (+6.1), BERTScore (+0.017), and FactCC (+3.8). Ablation studies show each module contributes meaningfully to both interpretability and accuracy, while low-resource experiments confirm ETS maintains high performance with limited fine-tuning. By explicitly modeling the elimination process, ETS enhances trust, transparency, and pedagogical value—making it particularly suitable for AI tutors, classroom tools, and exam preparation systems. Our work bridges cognitive strategies and LLM capabilities, offering a scalable path toward explainable and self-aware educational AI.

1. Introduction

Multiple-choice questions (MCQs) are ubiquitous in education for assessments and practice, covering domains from science and math to language comprehension (Haladyna, 2004; Schleicher, 2019). The rise of large language models (LLMs) has led to their adoption in educational question-answering (QA) systems, with mixed outcomes. On one hand, state-of-the-art LLMs can attain high accuracy on standard benchmarks, sometimes rivaling human test-takers (Achiam et al., 2023). On the other hand, these models often lack explainability and proper confidence calibration in their answers, which are critical in learning environments (Doshi-Velez & Kim, 2017; Guo et al., 2017). A student using an LLM-based tutor must not only receive correct answers but also understand the reasoning behind those answers and gauge the certainty of the model's response (Lipton, 2018). Simply providing an answer with no rationale or an overconfident incorrect answer can mislead learners, reducing the educational value of the system (Moore, 2024).

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Human test-takers frequently use the process of elimination to solve MCQs – they first discard obviously wrong options, then focus on the remaining options before selecting the best answer (Coderre et al., 2004). This step-by-step elimination strategy improves accuracy and gives insight into why certain options are wrong, thereby enhancing understanding (Kojima et al., 2022). Emulating this human-like strategy in LLMs could address the interpretability and reliability gaps (Zelikman et al., 2022). Recent work has suggested that prompting LLMs to “think step by step” can improve complex reasoning (Wei et al., 2022). However, vanilla chain-of-thought prompting often produces a single narrative reasoning for one final answer, without explicitly considering each alternative option (Lions et al., 2022). This can lead to selection bias and brittle reasoning — the model might fixate on a tempting option and concoct a reasoning that ignores other choices, sometimes even favoring a particular option position due to idiosyncratic biases (Zhang et al., 2025; Zheng et al., 2023). We posit that forcing the model to examine each option independently with rationale and confidence, as a human would, can yield more robust performance on MCQs.

This paper addresses the challenge of using LLMs to solve multiple-choice questions through elimination-based reasoning. We assume access to powerful pre-trained LLMs (e.g., Qwen, LLaMA, Yi, Baichuan, Phi, Gemma) which we treat as black-box solvers. We cannot alter their internal architecture, but we can prompt them or lightly fine-tune them to execute specific reasoning patterns. The core problem is to design a framework that orchestrates an LLM’s behavior to mimic elimination-before-selection, thereby improving both the accuracy of the final answer and the interpretability of the process. We introduce a modular framework called Eliminate-Then-Select (ETS) that decomposes the MCQ-solving process into multiple stages (Yang, Song et al., 2023). The system first generates option-specific rationales and confidence scores (Module A), then eliminates unlikely options (Module B), possibly iterates this elimination or triggers other reasoning strategies via a controller (Module C), and finally produces a calibrated answer with explanation (Module D). All modules are designed to be lightweight wrappers around the base LLM or simple supplementary classifiers, ensuring we do not require training a new large model from scratch. This design allows plug-and-play usage with any pre-trained LLM and supports extensibility — each module can be improved or replaced independently and ablated to assess its contribution.

In a typical run, given a question and multiple options, Module A queries the LLM to obtain a chain-of-thought (CoT) reasoning for each option along with a confidence estimate of that option being correct. For example, for a biology question with options A–D, the LLM might produce a few sentences evaluating option A (e.g. “Option A: X is true, but it does not account for Y, so it is likely incorrect”) and similarly for B, C, D, each ending with a confidence score. Module B then eliminates options whose confidence falls below a certain threshold or that a trained elimination classifier flags as likely incorrect. Suppose options A and C are eliminated in this step; the system is left with B and D. Module C (strategy controller) examines the situation: if more than one option remains with comparable confidence, it can prompt the LLM to “self-debate” – i.e. directly compare the merits of the remaining options or generate a new joint rationale to differentiate them. Alternatively, if one option’s confidence is overwhelmingly high, Module C may choose to skip directly to the final answer. In some cases, Module C can also decide to perform another round of elimination (back to Module A with updated context that certain options are removed) especially if many options remain. Finally, Module D generates the final answer along with an explanation synthesized from the retained reasoning chains, and calibrates the answer’s confidence. The calibration adjusts the model’s reported confidence to better reflect the true probability of correctness, using either a simple threshold rule or a learned calibration model, and ensures that the provided explanation covers both why the chosen answer is correct and why the eliminated options are wrong (making the reasoning self-consistent and complete) (Jiang et al., 2021).

The primary objective of this study is to improve the accuracy, interpretability, and confidence calibration of large language models (LLMs) in solving multiple-choice questions (MCQs) in educational settings. While existing approaches often treat MCQ options equally and select answers directly, they lack the human-like strategy of step-by-step elimination and do not provide structured justifications for both correct and incorrect options. This gap leads to brittle reasoning, limited trust, and reduced pedagogical utility. To address these challenges, we propose a modular framework named Eliminate-Then-Select (ETS) that mimics human test-taking behavior through a sequential elimination process. The key research objectives are as follows: **(O1)** To design a plug-and-play architecture that integrates chain-of-thought reasoning, option-level confidence estimation, and elimination strategies for improved decision-making. **(O2)** To implement a dynamic strategy controller that adapts reasoning paths via iterative elimination and self-debate, depending on confidence levels. **(O3)** To calibrate model confidence in alignment with actual correctness likelihood, making the system more reliable and cautious in high-stakes educational contexts. **(O4)** To empirically evaluate the effectiveness of the proposed framework across diverse educational QA benchmarks and under low-resource conditions. **(O5)** To enhance explainability by generating comprehensive final justifications that explicitly address both chosen and eliminated options. Through these objectives, our work aims to bridge the gap between raw accuracy and educational usability of LLMs, paving the way for interpretable and trustworthy AI tutoring systems.

Finally, we summarize our main contributions as follows:

1. We propose the Elimination before Selection (ETS) framework, a comprehensive approach to solving multiple-choice questions (MCQs). ETS uniquely integrates per-option chain-of-thought reasoning, explicit option elimination, adaptive strategy selection (including iterative elimination and self-debate), and joint answer-explanation calibration into a cohesive system. This unified architecture significantly extends and synthesizes existing ideas, enhancing robustness and practical effectiveness.
2. ETS systematically guides large language models (LLMs) to reason explicitly about each option, compelling them to articulate rationales for both correct and incorrect answers. This structured approach not only substantially improves answer accuracy but also provides high-quality explanations, clarifying common misconceptions and offering valuable interpretative insights for educational contexts.

3. Our framework incorporates an innovative joint calibration module that aligns the model's expressed confidence with its actual correctness likelihood, significantly reducing calibration errors. This calibration ensures that generated responses are both reliable and cautious, making the system particularly suitable for sensitive educational and high-stakes scenarios.
4. We rigorously evaluate ETS across six diverse educational MCQ benchmarks, consistently demonstrating superior performance compared to state-of-the-art baselines. Furthermore, we conduct detailed ablation studies to quantify each module's impact, validate ETS's robustness in low-resource fine-tuning scenarios. These analyses yield deeper understanding of model behaviors and validate the efficacy of elimination-based reasoning strategies.

Our findings suggest that combining human-inspired test-taking strategies with LLM reasoning yields a more trustworthy and pedagogically useful QA system. The improved interpretability and calibrated confidence can increase user trust and make the model's knowledge more accessible to learners. We envision this approach being applied to intelligent tutoring systems, exam preparation tools, and classroom support, where an AI assistant can show its work, explain alternatives, and know its limits — much like a skilled human tutor.

2. Related work

2.1. LLMs in educational question answering

The application of large language models to educational question answering (QA) has attracted growing interest as LLM capabilities have expanded. Early neural models for educational QA often required fine-tuning on domain-specific data (Sun et al., 2018), but modern LLMs like GPT-3 and GPT-4 have demonstrated strong zero-shot and few-shot performance on tasks such as standardized exams and curriculum questions (Achiam et al., 2023). For instance, GPT-4 reportedly achieved near-human scores on the Multitask Language Understanding (MMLU) benchmark, which includes many academic multiple-choice questions (Hendrycks et al., 2020). Likewise, LLMs have been evaluated on school science questions (ARC) and reading comprehension exams (RACE), showing substantial improvements over earlier architectures (Clark et al., 2018; Lai et al., 2017).

Despite these successes, challenges remain when deploying LLMs in educational settings. One challenge is domain adaptation: specialized exam questions (e.g., medical or legal exams) may contain terminology or styles of reasoning that general-purpose LLMs struggle with (Dash et al., 2023). This motivates techniques like prompt tuning or few-shot learning to adapt LLMs to specific educational domains. Another challenge is trust and interpretability. Unlike a traditional QA system that might retrieve facts, an LLM often generates answers end-to-end, which could include reasoning errors or unsupported claims. In education, it is crucial to have explainable outputs — e.g., a student using an automated tutor should be able to see why an answer is correct. Prior work has explored generating explanations or rationales alongside answers to increase transparency. However, producing a coherent and correct explanation is non-trivial; LLMs might give a right answer for the wrong reasons, or vice versa. Ensuring that the explanation truly justifies the answer is an ongoing area of research, overlapping with efforts in self-explaining models and faithful reasoning generation (Lyu et al., 2023).

Lastly, the evaluation of LLMs on educational QA often goes beyond simple accuracy. Metrics like partial credit, reasoning correctness, and even human student learning outcomes have been considered. Our work aligns with this trend by not only measuring accuracy but also analyzing the quality of reasoning (explanation correctness) and calibration of confidence — properties that are highly relevant in education.

2.2. Strategies for solving multiple-choice questions

Solving multiple-choice questions has been a long-standing problem in QA research, with strategies evolving alongside model capabilities. Traditional methods either treated MCQs as classification (select the correct option label) or as generative (output the text of the correct answer) tasks. Classic neural approaches often used an elimination style implicitly by scoring each option (e.g., via an entailment model or reading comprehension model) and choosing the highest scoring option (Zhu et al., 2016). With LLMs, a new paradigm emerged: chain-of-thought prompting, which encourages the model to reason step-by-step rather than immediately outputting an answer. In the context of MCQs, chain-of-thought (CoT) prompting might yield a paragraph of reasoning that ends in a final answer choice. While CoT often improves accuracy by reducing reasoning errors, it does not inherently implement option elimination — the model might only focus on one option in its reasoning path, missing the chance to explicitly rule out alternatives.

Researchers have thus begun exploring process-of-elimination (PoE) techniques with LLMs. One approach is to prompt the LLM to eliminate options directly. For example, a prompt might say: “List which options are definitely incorrect and why, then choose the answer from the remaining”. Some studies (Chen, 2024) tried this incorrect option identification approach, but found that current LLMs struggle to reliably identify wrong answers in one go — often because explaining why something is incorrect may require as much reasoning as finding the correct answer, and models can get confused or produce inconsistent eliminations. Another line of work scores each option separately with the LLM. In this scoring approach, the model is asked (either via likelihood or an explicit question) how likely each option is correct, possibly using chain-of-thought for each, and then eliminates low-scoring options (Ma & Du, 2023). This was shown to improve performance on some reasoning tasks by effectively narrowing the model's focus. However, it comes at the cost of multiple LLM forward passes (one per option) and can still fail if the model's scoring is unreliable.

Recent innovations have looked at making elimination more efficient and robust. Sequential elimination (Zhu et al., 2025) is a strategy where the model eliminates one option at a time in a multi-turn interaction. For instance, the model might first choose

the single worst option to eliminate, then reconsider the question with that option removed, and repeat. This sequential PoE has two benefits: (1) it mirrors human test-taking more closely (eliminating stepwise), and (2) it can enhance the model's reasoning by focusing its attention progressively (Zhu et al., 2025). However, sequential approaches raise the question of how to control the sequence: how many options to eliminate, when to stop, etc. That motivates a strategy controller in our framework (Module C), which can decide between one-shot elimination vs. multi-step elimination vs. other tactics like prompting the model to compare two finalists (a form of self-debate). The idea of self-debate or letting a model argue for and against options has been explored conceptually in past work on AI debate frameworks, and more recently, some prompting techniques get an LLM to critique or counter-argue its initial answers to improve correctness (Madaan et al., 2023). We incorporate a simpler version of this idea: if two options remain and the model's confidence is split, we prompt it to generate a comparative reasoning (e.g., "Discuss why option B might be better than D or vice versa"), treating it as an internal debate to resolve uncertainty.

In addition to elimination strategies, researchers have examined self-consistency methods for MCQs (Wang et al., 2022). Self-consistency involves generating multiple independent reasoning paths (with an LLM sampling different chains-of-thought) and then taking a majority vote of the answers. This can often boost accuracy by reducing random reasoning errors. However, it is computationally expensive and does not guarantee an explanation that covers wrong options. Our method instead focuses on one thorough reasoning per option, which serves a similar ensemble purpose (each option is evaluated separately) and results in a built-in explanation for all options. Moreover, some works identified specific biases and robustness issues in LLM multiple-choice answering, such as the tendency to prefer certain option positions or to be influenced by irrelevant context (Zheng et al., 2023). Debiasing techniques (e.g., shuffling options or calibrating with permuted inputs) have been proposed. Our framework could integrate such techniques in Module B or Module D if needed (for example, the elimination classifier could account for option position bias), but our main contributions are orthogonal to those and focus on reasoning strategy.

2.3. Explainable reasoning and confidence calibration

Explainability and calibration are key to making AI systems trustworthy, especially in education where understanding the reasoning process is as important as the answer. Explainable QA has been studied by requiring models to output human-readable justifications or by training models to highlight supporting evidence for their answers (Das & Rad, 2020). With LLMs, explainability often comes in the form of a chain-of-thought – a narrated reasoning (Wei et al., 2022). However, a known issue is faithfulness: the explanation generated by the model might not reflect the true cause of its answer, and sometimes models can produce a convincing explanation for a wrong answer ("falling for their own false reasoning") (Lyu et al., 2023). To mitigate this, techniques like verification of reasoning have been proposed, where a separate model or heuristic checks if the explanation indeed supports the answer (Fayyaz et al., 2024). In our framework, the answer-explanation calibrator (Module D) plays a role in ensuring consistency: it can verify that the final explanation addresses all eliminated options and that the reasoning for the chosen answer contains no obvious gaps or contradictions. This can be partially automated by rules (e.g., each eliminated option must be mentioned with a rationale) and by using the LLM itself to double-check (e.g., asking the LLM or another smaller model "Does this explanation correctly justify why the answer is correct and others are wrong?").

Confidence calibration refers to aligning the model's confidence score with the actual probability of correctness. A perfectly calibrated model will, for example, have its "80% confident" answers be correct 80% of the time. Calibration is crucial for AI tutors: if a model is unsure, it could signal the student or refrain from giving a definitive answer, whereas a highly confident answer can be presented assertively. Neural models, including LLMs, are often poorly calibrated out-of-the-box (Guo et al., 2017). For instance, an LLM might assign very high probability to one option even when it has only a slightly better chance of being correct than others, or it might be overconfident in cases it actually gets wrong. Prior research has explored methods to calibrate QA systems, such as temperature scaling, Platt scaling, or more advanced methods utilizing model uncertainty measures (Kopetzki et al., 2021). In multiple-choice QA, one can derive confidence from the model's predicted distribution over options or from the logits if available, then apply calibration techniques (Adepoju et al., 2024). With black-box LLMs that generate text, an alternative is to have the model express its confidence in words or on a numeric scale, though models may still misestimate their certainty (Desai & Durrett, 2020).

Our framework explicitly incorporates confidence estimation at multiple points: Module A produces a confidence per option (which can be seen as intermediate calibration), and Module D produces a final answer confidence. We leverage simple yet effective calibration approaches: for example, we might fine-tune a sigmoid function on a validation set to map the raw LLM confidence scores to calibrated probabilities (Guo et al., 2017), or use the agreement between different reasoning strategies as a confidence signal (if direct answer and elimination reasoning agree, confidence should increase, as noted by Tao et al. (2024)). Additionally, we evaluate calibration quality using metrics like Expected Calibration Error (ECE) (Guo et al., 2017) to verify that our joint answer-explanation outputs are more trustworthy. By combining explainable reasoning with calibrated confidence, the system can provide users with not just an answer and explanation, but also an indication of how reliable that answer is – a critical feature for deploying AI in real-world classrooms.

3. Methodology

3.1. System architecture

Our proposed Eliminate-Then-Select (ETS) system comprises four interconnected modules (A–D) that together implement an elimination-style reasoning process on top of a base LLM. Fig. 1 provides a schematic of the architecture. At a high level, the system takes as input a multiple-choice question (consisting of a question prompt and a set of options) and produces as output a final answer choice along with an explanation and a confidence score. The workflow is as follows:

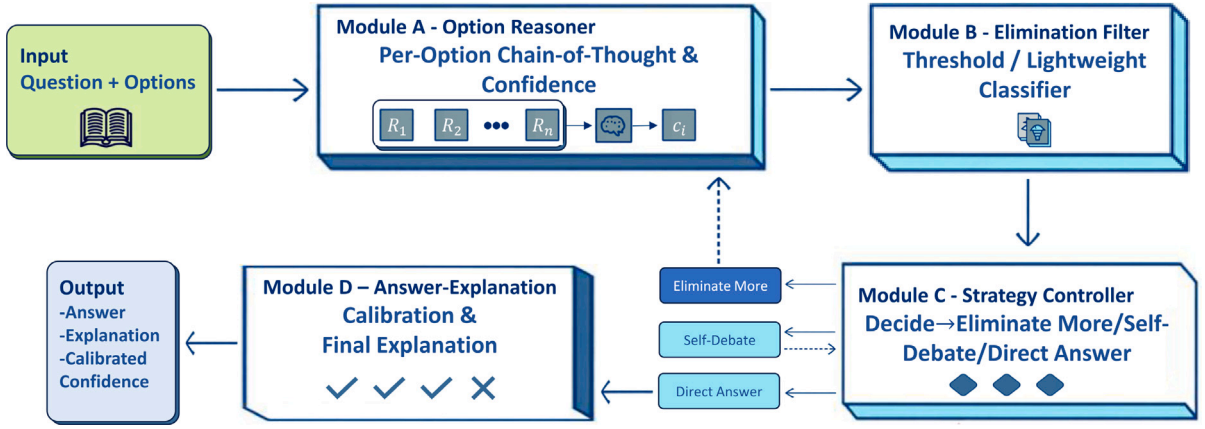


Fig. 1. The main idea of our ETS system.

- Module A (Option Reasoner):** For each option, query the LLM to generate a chain-of-thought rationale assessing that option's correctness, and produce a numerical confidence estimate for the option (e.g., a probability or score). This module operates in parallel for all options (or sequentially iterating through options) and treats the LLM as a black box that can generate text conditioned on a prompt.
- Module B (Elimination Filter):** Based on the rationales and confidence scores from Module A, eliminate one or more options deemed likely to be incorrect. This can be done via a simple threshold on the confidence (e.g., eliminate options with confidence below 0.2) or via a trained lightweight classifier that reads the option's rationale and predicts if the option should be eliminated. The output is a subset of options that remain "active" after elimination.
- Module C (Strategy Controller):** Decide the next action depending on the number and confidence of remaining options. If multiple options remain with no clear winner, Module C may choose to iterate (go back to Module A/B for another round with updated context) or invoke a special reasoning strategy like a pairwise comparison (self-debate) between top contenders. If only one option remains or one is far more confident than others, Module C can proceed to final answer output. This module encapsulates the decision logic or policy (which could be rule-based or learned) that controls the reasoning flow dynamically.
- Module D (Answer and Explanation Calibrator):** Produce the final answer choice along with a consolidated explanation and calibrate the confidence. The explanation is constructed by combining the reasoning chains for the chosen answer and key points from eliminated options' rationales, thereby justifying why the answer is correct and others are not. The confidence for the final answer is adjusted (calibrated), for example by scaling it or mapping it through a calibration curve, to ensure it reflects actual correctness likelihood as observed on validation data. This module may also perform a final check for consistency, potentially using the LLM to verify the explanation, before outputting the result.

Each module can be ablated or replaced independently, which not only aids in experimental analysis (we can turn off elimination or the strategy controller to see the effect) but also provides flexibility for different use-cases. For instance, if computational resources are limited, one might simplify Module C to always do one elimination round then answer, whereas in a critical setting requiring maximum accuracy, Module C could be configured to attempt multiple elimination rounds or debates until a threshold of confidence is reached.

We next describe each module in detail, including formal definitions and pseudocode. Table 1 summarizes the notation used throughout the Method section.

3.2. Module A: Per-option chain-of-thought reasoning with confidence estimation

Module A's role is to independently evaluate each answer option through a chain-of-thought and to quantify the likelihood that the option is correct. Formally, consider a multiple-choice question consisting of a prompt Q and a set of N options $\{O_1, O_2, \dots, O_N\}$. For each option $i \in \{1, \dots, N\}$, Module A generates a reasoning trace R_i (a sequence of natural language statements) and a confidence score $c_i \in [0, 1]$ indicating the model's estimated probability that O_i is the correct answer to Q .

To obtain R_i and c_i , we utilize the base LLM in a prompted manner. We design a prompt template that injects the question Q and the candidate option O_i into a prompt instructing the model to reason about the option's correctness. For example, a prompt could be: "Question: Q \n Option i: O_i \n Reasoning:". The LLM, upon receiving this prompt, is expected to continue by writing a few sentences analyzing O_i . We may also append an instruction like "Evaluate if this option is correct or not and explain". to encourage critical analysis. The LLM's completion to this prompt is captured as the raw reasoning text for option i .

From the raw reasoning, we need to extract a confidence score c_i . We explore two methods for this:

Table 1
Notation for Module A (Option Reasoner).

Symbol	Description
Q	The question text (prompt of the MCQ).
O_i	The i th answer option text for the question.
R_i	Chain-of-thought reasoning generated by the LLM for option O_i .
c_i	Confidence score (probability) that option O_i is correct, as estimated by Module A.
LLM	The large language model used for generation.
f	Mapping from LLM output (text or logits) to confidence score.

- **Implicit Confidence from Text:** We instruct the LLM to end its reasoning with a statement of likelihood, for instance: “Conclusion: This option is likely incorrect (10% confidence)”. If the model outputs a percentage or qualitative likelihood (certainly correct, probably incorrect, etc.), we map that to a numeric score. We have to standardize this across outputs (e.g., if it says “10% confidence”, then $c_i = 0.1$; if it says “likely correct”, we might assign $c_i = 0.8$, etc. based on a predefined mapping).
- **Explicit Scoring via Logits:** If the LLM allows probability estimates (some LLM APIs or models can return log probabilities for next token predictions), another approach is to score the option by turning the task into a fill-in-the-blank. For instance, we could prompt: “Q\nOptions: (A) ... (B) ... (C) ... (D) ...\nAnswer:” and examine the probability the model assigns to each option letter as the next token. This yields a probability distribution over options in one forward pass. In this case, c_i can be the normalized probability of option i according to the model’s next-token probabilities. However, this method treats the LLM like a classifier rather than generating a rationale, so we primarily use it in analysis or in Module B’s elimination strategy (as a complement).

In our implementation, we favor the implicit confidence from text approach for its flexibility and interpretability — the model explains its reasoning and “self-assesses” its confidence.

We denote the generated chain-of-thought for option i as

$$R_i = [r_{i1}, r_{i2}, \dots, r_{iT_i}], \quad (1)$$

where r_{ij} are the tokens or words in the explanation (with length T_i). The confidence c_i can be seen as the output of a function f applied to the reasoning:

$$c_i = f(R_i, Q, O_i). \quad (2)$$

In the case of the model providing an explicit percentage, f is just parsing that value. If we rely on the model’s final token probabilities, f would extract the probability of a special token corresponding to correctness. For theoretical formulation, we can assume

$$c_i = P(\text{option } i \text{ is correct} \mid Q, \text{LLM reasoning}). \quad (3)$$

Let Q be a question and $O_1, \dots, O_N$ be the answer options. Module A produces a set of rationale-confidence pairs $s\{(R_i, c_i)\}_{i=1}^N$. The rationale R_i is a sequence of text tokens representing the LLM’s reasoning for option O_i . The confidence c_i is a real number in $[0,1]$ representing the probability that O_i is correct, as estimated by the model. Key variables for Module A are summarized in Table 1. Notably, Q (question), O_i (option text), R_i (reasoning for option i), c_i (confidence for option i), and any internal prompt templates or LLM parameters (which we denote abstractly, since the LLM is a black box with fixed parameters).

The pseudocode below (Algorithm 1) outlines the operation of Module A for a single question. It assumes we have access to a function $LLM_{generate}(prompt)$ that returns the model’s completion given a text prompt.

Algorithm 1 Module A : Per-Option Reasoning and Confidence

Input: Question Q , Options $\{O_1, \dots, O_N\}$, Prompt template $TEMP_L$.

Output: Rationales $\{R_i\}$ and confidences $\{c_i\}$ for each option.

```

1: for  $i = 1$  to  $N$  do
2:   Construct prompt for option  $i$ :
3:    $prompt_i \leftarrow \text{format}(TEMP_L, \text{question} = Q, \text{option} = O_i)$ 
4:   Get LLM-generated reasoning for option  $i$ :
5:    $raw_{output_i} \leftarrow LLM_{generate}(prompt_i)$ 
6:   Split  $raw_{output_i}$  into rationale text and confidence phrase:
7:    $R_i, conf_{text_i} \leftarrow \text{parse}_{rationale \text{ and } confidence}(raw_{output_i})$ 
8:   Convert confidence text to numeric score:
9:    $c_i \leftarrow \text{parse}_{confidence}(conf_{text_i})$ 
10: return  $\{R_1, \dots, R_N\}, \{c_1, \dots, c_N\}$ 

```

In lines 3–5, we prompt the LLM for each option’s analysis. The *parse_rationale_and_confidence* function (line 7) depends on how we structure the LLM’s output. For example, we might enforce a delimiter like the word “Conclusion:” in the prompt template such that everything after “Conclusion:” is a confidence statement. The pseudocode treats this parsing abstractly. Finally, we parse that confidence into a numeric value (line 9). After this loop, we have an initial reasoning and confidence for every option.

Module A essentially converts the question into a set of annotated options: each option now has a justification and a score. This is the basis for the elimination step next. It is worth noting that Module A can be either zero-shot (no additional training, just prompting) or few-shot (we could include a few examples of the reasoning format in the prompt to guide the LLM). In our experiments, we test both zero-shot and few-shot versions of Module A.

3.3. Module B: Threshold-based option elimination filter

Module B takes the set of options with their rationales and confidence scores from Module A and filters out the options considered unlikely to be correct. The goal is to mimic the human practice of crossing out wrong answers, thereby reducing the pool of contenders. We implement Module B in two variants: (1) a simple threshold rule on c_i , and (2) a trained classifier that reads R_i to decide elimination.

- **Threshold Rule:** We choose a confidence threshold τ . Any option with $c_i < \tau$ is eliminated. The threshold τ could be fixed (e.g., 0.5) or adaptively chosen based on the distribution of c_i for a particular question (e.g., eliminate all but the top 2 confidences, or eliminate those more than 0.2 below the top score). A fixed threshold has the advantage of simplicity and can be tuned on a validation set to optimize accuracy. For instance, if set too high, we might eliminate the correct answer mistakenly; too low and we eliminate nothing. We found that a moderate τ (around 0.3–0.5 in probability) works well in practice, often eliminating about half the options initially.
- **Classifier-Based:** In this approach, we train a small binary classifier g that takes as input the text of an option’s rationale R_i (and possibly the question Q and option Q_i for context) and outputs 1 if the option should be eliminated (i.e., is predicted incorrect) or 0 if it should be kept. The rationale behind using a classifier is that certain patterns in the chain-of-thought might indicate confusion or incorrectness. For example, if R_i contains language like “I am not sure” or flawed logic, the classifier might learn to flag that. We train g on a small dataset of rationales labeled by correctness (which we can obtain by running Module A on some training questions where the true correct answer is known: rationales for correct options are positive examples to keep, rationales for wrong options are examples to eliminate). This classifier could be as simple as a logistic regression on embedding features or a tiny transformer on the text, given our desire to keep modules lightweight.

Regardless of method, Module B outputs an elimination mask: a set of binary indicators

$$E = \{e_1, \dots, e_N\} \quad (4)$$

where $e_i = 1$ means option i is eliminated (incorrect) and $e_i = 0$ means option i remains candidate. We also define the set

$$S = \{i \mid e_i = 0\} \quad (5)$$

as the indices of surviving options after elimination.

Formal Definition: Given option confidences $c_1, \dots, c_N$ and rationales $R_1, \dots, R_N$, Module B determines an elimination decision $e_i \in \{0, 1\}$ for each option. This can be defined by

$$e_i = \mathbb{I}[h(O_i, R_i, Q) < 0] \quad (6)$$

where h is a scoring function that yields positive values for likely correct options and negative for incorrect. In the threshold case, h could be simply

$$h = c_i - \tau (\text{so } e_i = 1 \text{ if } c_i < \tau). \quad (7)$$

In the classifier case, h might be the classifier’s output logit for option correctness. We then have the remaining set

$$S = \{i : e_i = 0\}. \quad (8)$$

Typically, we expect S to be significantly smaller than N . For example, if $N = 4$ (A through D), Module B might eliminate 2 or 3 options, leaving 1 or 2 finalists.

Additional notation for Module B is listed in Table 2. Algorithm 2 shows pseudocode for the threshold version of Module B (the classifier version would replace the condition with a classifier prediction).

If using a classifier instead, line 3 would be: “if $g(Q, O_i, R_i) = \text{“keep”}$ then...”. In practice, we might also enforce that at least one option remains (since it is possible all $c_i < \tau$). If the elimination rule eliminates everything (i.e., S becomes empty), we fallback to keeping the top one or two options by confidence — this ensures the pipeline has something to work with in Module C. Such a fallback is rarely needed if τ is tuned sensibly.

Module B thus produces a pruned set of options. The eliminated options’ rationales R_i are not discarded; they may be used later for the explanation or for a potential reconsideration if the strategy decides to second-guess elimination. However, at this point in the pipeline, we proceed as if those options are no longer in contention.

Table 2
Notation for Module B (Elimination Filter).

Symbol	Description
τ	Confidence threshold for elimination (if using threshold rule).
e_i	Elimination indicator for option i (1 = eliminated, 0 = kept).
S	Set of indices of options that survive elimination.
g or h	Classifier or scoring function used to decide elimination.

3.4. Module C: Strategy controller

Module C is the decision-making hub of the framework. Based on the current state of reasoning (confidences and remaining options after Module B, as well as possibly the content of their rationales), the strategy controller decides what to do next. It ensures that the system can dynamically adapt, rather than following a rigid one-pass elimination. We design Module C to handle three main strategies: Eliminate More, Self-Debate, or Direct Answer.

- **Eliminate More:** If after one round of Module A and B there are still multiple options and none has extremely high confidence, the controller can decide to perform another elimination round. This means we loop back to Module A, but now we condition on the knowledge that some options have been eliminated. Practically, we can prompt the LLM with an updated question that explicitly removes or ignores certain options (e.g., “Given that options A and C are definitely wrong, now consider option B...”). Alternatively, we could run Module A fresh on the reduced set of options to get refined rationales. Sequential elimination as studied by [Zhu et al. \(2025\)](#) often improves reasoning focus, so Module C may choose to eliminate options one by one in multiple iterations if needed. We typically limit the number of elimination iterations (say, maximum 2 or 3 rounds) to avoid excessive computation and potential looping if the model oscillates.
- **Self-Debate (Pairwise Comparison):** If exactly two options remain (or if after elimination the top two options have close confidence scores), the controller can invoke a self-debate strategy. Here we prompt the LLM in a different way: instead of analyzing one option in isolation, we ask it to compare or debate the pros and cons of the two remaining options. For example: “Only options B and D remain. Compare these two options and explain which is more likely correct”. The LLM will then generate a comparative analysis. This often helps resolve subtle distinctions that the independent rationales might not have conclusively addressed. We have found that an LLM’s answer after a direct comparison can overturn initial confidence if it realizes a detail was missed. The output of this self-debate can either directly yield a final answer (if the analysis clearly favors one option), or it could even output something like “Option B is more likely correct” which we then take as the answer. We treat the self-debate essentially as a special case of Module A reasoning, but for a pair of options together.
- **Direct Answer:** If after elimination there is a single option left, or one option’s confidence is far above the rest (above a second threshold τ_{answer} , e.g., 0.9), the controller will choose to finalize the answer without further elimination or comparison. In this case, the system is confident enough to proceed to Module D. Direct answer could also be the default if for some reason Module B did not eliminate anything (then we might just take the highest confidence option as answer, but with caution and calibration).

The strategy controller can be implemented with simple rules or a learned policy. In our framework, we use a rule-based approach for clarity and predictability. The rules can be summarized as:

1. If $|S| > 2$ (more than 2 options remain) and not at iteration limit: Eliminate More. (Go back to Module A with those options).
2. Else if $|S| = 2$: Self-Debate.
3. Else if $|S| = 1$ or max confidence $\max_{i \in S} c_i$ is above τ_{answer} : Direct Answer.
4. Else: (This covers $|S| = 0$, which should not happen due to fallback, or any other odd case) take the top option as answer anyway (with low confidence).

Algorithm 2 Module B: Option Elimination Filter (Threshold Version)

Input: Options $\{O_i\}$, Confidence scores $\{c_i\}$, Threshold τ

Output: Elimination mask $\{e_i\}$, Remaining options S

```

1: Initialize empty set  $S$ 
2: for  $i = 1$  to  $N$  do
3:   if  $c_i \geq \tau$  then
4:      $e_i \leftarrow 0$  # keep option  $i$ 
5:     add  $i$  to  $S$ 
6:   else
7:      $e_i \leftarrow 1$  # eliminate option  $i$ 
8: return  $\{e_1, \dots, e_N\}, S$ 

```

Table 3
Notation for Module C (Strategy Controller).

Symbol	Description
S	Set of remaining option indices after elimination.
τ_{answer}	Confidence threshold to decide direct answer output.
Action	Chosen strategy (eliminate, debate, or answer).
Iter	Current elimination iteration count (to enforce a max limit).

These rules ensure the process ends in a finite number of steps. In the worst case, if elimination keeps failing to reduce options (e.g., model is equally unsure about all), we might set an iteration cap and then force a decision by direct answer from top confidence.

Module C defines a policy $\pi(S, \{c_i\}_{i \in S}, \text{context}) \rightarrow \text{EliminateMore}, \text{SelfDebate}, \text{DirectAnswer}$. The context may include the entire set of rationales or any other metadata. Our rule-based policy can be represented as a set of conditions as described above. One could also imagine a learned policy where, for example, a small neural net takes the vector of remaining confidences and perhaps some embedding of rationales to decide the next step, but we found rules sufficient.

Notation for Module C is straightforward: S (remaining options set, from Module B), and thresholds or conditions as defined. We introduce τ_{answer} as a high-confidence threshold for directly answering. Also, an iteration counter could be tracked (see Table 3).

Algorithm 3 demonstrates the logic of Module C in a simplified form.

Algorithm 3 Module C - Strategy Controller (Rule-Based)

Input: Remaining option set S , confidences $\{c_i \text{ for } i \in S\}$, iteration count k

Output: Next action (*eliminate_{more}*, *self_{debate}*, *direct_{answer}*)

```

1: if  $|S| > 2$  and  $k < K_{\max}$  then
2:   action  $\leftarrow$  "eliminatemore"
3: else if  $|S| = 2$  then
4:   action  $\leftarrow$  "selfdebate"
5: else
6:   #  $|S| = 1$ , or we have reached a stopping condition
7:   action  $\leftarrow$  "directanswer"
8: return action

```

Here $K_{\max}$ is a predefined maximum number of elimination rounds (for example, 2). If we wanted to incorporate the confidence threshold τ_{answer} , we could modify line 3–5 logic: e.g., if $|S| > 1$ but one option has $c_i > \tau_{\text{answer}}$, we might short-circuit to direct answer rather than debating.

When the controller chooses “eliminate_{more}”, the system will loop back: run Module A again on the remaining options S (possibly with a prompt indicating which options are gone) and Module B again to further prune. When “self_{debate}” is chosen, we create a special prompt for the LLM to compare the two options in S . This could be seen as a modified Module A that takes two options at once. The outcome of self_{debate} is typically a new rationale and potentially updated confidences for those two (often one will emerge as clearly preferred). Finally, “direct_{answer}” means proceed to Module D with the current best option.

3.5. Module D: Answer-explanation joint calibrator

Module D produces the final answer, an explanation, and a calibrated confidence. It is the last step of the pipeline, consolidating information from previous modules. This module addresses two aspects: (1) Joint answer-explanation generation and (2) Confidence calibration.

For the answer and explanation, a straightforward approach is to take the surviving rationale(s) from the prior steps and concatenate them into a coherent explanation. For instance, if option B is the final answer, we might take R_B (the reasoning for B) and also append sentences from the rationales of eliminated options that explicitly say why those were wrong. We can prompt the LLM to perform this synthesis for better fluency: e.g., “Based on the analysis, the correct answer is B. Explain why B is correct and why the other options are incorrect”. This prompt can use the content from R_i for eliminated options $i \notin S$ as reference (we may even feed those rationales into the prompt). The LLM then generates a nicely formatted explanation paragraph. In many cases, R_B might already contain a lot of this, but doing this final rephrasing ensures the explanation is readable and consolidated rather than a disjoint collection of thoughts. We call the final explanation F_{final} .

The confidence output for the final answer, denote it C_{final} , should ideally represent the model’s true probability of being correct. We obtain a raw confidence from the process — if only one option remains, we likely have a confidence for it from Module A’s last run or from the self-debate step. However, this raw confidence may be biased. To calibrate it, we use a simple calibration mapping learned on a validation set. Specifically, we fit a logistic regression (Platt scaling [Platt 1999]) on the model’s raw confidence vs. ground truth correctness. In practice, since we might not have a huge training set for calibration, we could assume reliability of

Table 4
Notation for Module D (Answer & Calibration).

Symbol	Description
a^*	Index of the selected correct answer option (final output).
A_{final}	The final answer option text (same as O_{a^*}).
E_{final}	The final compiled explanation text.
C_{final}	Calibrated confidence score for the final answer.
$\text{cal}(\cdot)$	Calibration mapping function (e.g., logistic curve).

frequencies: e.g., if over many validation questions, whenever the model said 90% it was correct 80% of the time, we adjust 90% down to 80%. This yields a calibration curve $\text{cal}()$. We then set

$$C_{\text{final}} = \text{cal}(\max_{i \in S} c_i) \quad (9)$$

(the highest confidence among remaining, which corresponds to the answer's confidence). If no labeled data is available for calibration, an alternative is to use the heuristic that if the model went through elimination and debate and still is unsure (maybe two options were neck-and-neck), we output a lower confidence, whereas if it eliminated all but one easily, we output a high confidence. Our experiments in Section Experiments demonstrate that even a basic calibration significantly improves the correlation of C_{final} with actual accuracy.

Module D can also perform a final consistency check. We want to ensure E_{final} (the explanation) is aligned with the answer choice. One approach is to have a second pass where we ask the LLM (or another model) to verify: "Question, Answer, Explanation — is the explanation correct and does it support the answer?" If the verifier says no or is uncertain, we might mark the output as potentially unreliable. In our current framework, we simply note such cases in analysis but do not modify the answer. In future work (as we will discuss in Conclusion), such verification could be used to trigger the system to reconsider (e.g., try an alternate reasoning path if the explanation seems inconsistent).

Let a^* be the index of the final selected answer option (so the answer text is O_{a^*}). Module D produces

$$A_{\text{final}} = O_{a^*}, E_{\text{final}} \quad (10)$$

as a textual explanation, and C_{final} as a calibrated confidence. Formally,

$$C_{\text{final}} = \text{cal}(p_{a^*}) \quad (11)$$

where p_{a^*} is the raw probability for option a^* from the earlier step (e.g., c_{a^*} after last Module A) and $\text{cal} : [0, 1] \rightarrow [0, 1]$ is a calibration mapping (monotonic). The explanation E_{final} is a function of all rationales and elimination decisions:

$$E_{\text{final}} = g(\{R_i, e_i\}_{i=1}^N), \quad (12)$$

where g could be as simple as concatenation or involve an LLM-generated rephrasing.

Key variables are a^* (index of final answer), E_{final} (final explanation text), C_{final} (final confidence). Also $\text{cal}(\cdot)$ representing the calibration function (which could be defined by a couple of parameters if logistic, or a lookup table if non-parametric) (see Table 4).

Algorithm 4 illustrates Module D's procedure. In step 1–5, we identify which option to output as the answer (usually there will be exactly one in S by this stage; if two remain because Module C decided to directly answer even though two left, we just pick the higher confidence). Step 10 shows using the LLM to form the explanation — $\text{format}_{\text{explanation}}(\text{prompt})$ would craft a prompt like:

Algorithm 4 Module D: Answer-Explanation Calibration and Output

Input: Remaining option set S , rationales $\{R_i \text{ for } i \in S \text{ and } i \text{ eliminated}\}$, confidences $\{c_i\}$, calibration function $\text{cal}()$

Output: Final answer A_{final} , explanation E_{final} , calibrated confidence C_{final}

```

1: Determine final answer index:
2: if  $|S| = 1$  then
3:    $a^* \leftarrow$  the single element in  $S$ 
4: else
5:    $a^* \leftarrow \arg \max_{i \in S} c_i$  # choose option with highest confidence
6: # Generate final explanation by combining rationales
7: let  $\text{correct}_{\text{rationale}} \leftarrow R_{a^*}$ 
8: let  $\text{wrong}_{\text{rationales}} \leftarrow \{R_j : j \notin S\}$  # all eliminated options' reasoning
9:  $E_{\text{final}} \leftarrow \text{LLM}_{\text{generate}}(\text{format}_{\text{explanation}}(\text{prompt})(Q, O_{a^*}, \text{correct}_{\text{rationale}}, \text{wrong}_{\text{rationales}}))$ 
10: # Calibrate confidence
11:  $\text{raw}_{\text{conf}} \leftarrow c_{a^*}$  # latest confidence for chosen answer
12:  $C_{\text{final}} \leftarrow \text{cal}(\text{raw}_{\text{conf}})$ 
13: return  $O_{a^*}, E_{\text{final}}, C_{\text{final}}$ 

```

“Question: Q \n The correct answer is O_a . Explain why this is correct and why the other options are incorrect: \n” followed by perhaps listing the other options and their issues (we can insert some of the content from *wrong_rationales* for guidance). The LLM’s output to this prompt is our final explanation. Alternatively, we could assemble E_{final} without an LLM by template, but that might be less fluent.

Finally, we apply $cal()$ to the raw confidence (line 12). If no calibration is available, cal could be the identity function (making $C_{final} = raw_{conf}$), but we recommend at least using a simple heuristic calibration as described. With Module D, the pipeline is complete. The user (or evaluation metric) receives the answer choice, the explanation of reasoning, and a confidence level.

3.6. Silver reference construction for explanation evaluation

Several of the datasets used in our experiments (e.g., PIQA, RACE, OpenBookQA) do not contain human-written rationales that could serve as gold references for explanation quality metrics such as ROUGE or BERTScore. To enable a fair and consistent evaluation across datasets, we construct silver references using a standardized protocol:

- **Step 1: Input Preparation.** For each question, we provide the question stem, all candidate options, and the official correct answer to GPT-4. No gold explanations or external solutions are included in the input.
- **Step 2: Prompting Protocol.** We design a fixed prompt (Fig. 2) that instructs the model to (i) generate a concise support rationale for the correct option, and (ii) provide one elimination reason for each incorrect option. If a passage or supporting text is available in the dataset, the model is instructed to cite the exact sentence indices or spans used as evidence.
- **Step 3: Human Verification.** Two independent annotators verify each generated explanation for correctness and faithfulness to the provided question and evidence. Disagreements are adjudicated by a third reviewer.
- **Step 4: Usage in Evaluation.** These silver references are used only as target texts for automatic overlap-based metrics (ROUGE, BLEU, BERTScore). They are never used as model inputs during inference.

This silver reference construction allows us to compute explanation metrics consistently even on datasets without gold rationales, while keeping the process transparent and reproducible. In addition to these silver references, we introduce a new benchmark dataset with gold explanations curated from licensed exam materials, which serves as a complementary evaluation setting with human-authored rationales.

4. Experiments

We conduct a series of experiments to evaluate our proposed ETS framework on multiple benchmarks, to compare it with baseline methods, and to analyze its components. First, we describe the datasets and baseline setups. Then we present the main results, followed by ablation studies, low-resource fine-tuning results, and additional analyses on reasoning quality and calibration. All experiments use the same overall framework, with specific LLMs and parameters as described.

4.1. Datasets

In this study, we evaluate our proposed method comprehensively on six publicly available educational multiple-choice question datasets, each uniquely testing different reasoning capabilities and knowledge domains. These datasets were carefully selected to represent diverse reasoning challenges and knowledge-intensive tasks typically encountered in educational contexts. The datasets (Table 5) are summarized as follows:

- **Physical Interaction Question Answering (PIQA)** (Mihaylov et al., 2018): PIQA consists of multiple-choice questions focusing on physical commonsense reasoning. Each question provides two candidate solutions related to physical actions, such as everyday problem-solving scenarios. The primary evaluation metric for this dataset is accuracy, calculated as the percentage of correctly answered questions. PIQA challenges models to understand and apply commonsense knowledge about physical interactions. The dataset is publicly available at: https://huggingface.co/datasets/d0rj/piqa_ru.
- **OpenBookQA** (Mihaylov et al., 2018): OpenBookQA is an open-domain science question-answering dataset derived from elementary-level science concepts. Each question is accompanied by four possible answer choices. The questions require complex reasoning, often necessitating the integration of provided scientific facts (“open-book”) with external knowledge. The dataset is accessible at: <https://huggingface.co/datasets/allenai/openbookqa>.
- **Reading and Reasoning Ability Comprehension (RACE)** (Lai et al., 2017): RACE assesses reading comprehension and inferential reasoning abilities through multiple-choice questions extracted from language comprehension examinations. Each question has four candidate answers and demands nuanced understanding of given textual passages and inferential capabilities. Download link: <https://huggingface.co/datasets/ehovy/race>.
- **ScienceQA** (Lu et al., 2022): ScienceQA features multiple-choice science questions intended for elementary and middle school students, spanning biology, chemistry, and physics. Each question presents four possible answers, frequently accompanied by textual or visual contextual information (although our experiments utilize only textual components). The dataset particularly emphasizes scientific reasoning and concept comprehension. The primary evaluation criterion is accuracy. Dataset link: <https://github.com/lupantech/ScienceQA>.

Prompt 1: Prompt Templates for Silver Explanations

[Task]
 Given a multiple-choice question, its candidate options, a supporting passage (split into sentences and indexed), and the official correct option,
 (1) write a concise support rationale for the correct option that cites exact sentence indices from the passage, and
 (2) give one short elimination reason for each incorrect option, also citing sentence indices if relevant.

[Inputs]
 Question: QUESTION_TEXT
 Options:
 A. OPTION_A B. OPTION_B C. OPTION_C D. OPTION_D OfficialCorrectOption: CORRECT_LETTER // one of A/B/C/D
 Passage (sentence-indexed):
 [0] SENT_0 [1] SENT_1
 ...
 [n] SENT_n

[Style & Constraints] - Write in English.
 - Be specific and evidence-grounded. Cite sentence indices like [3], [7] where used.
 - Do NOT reveal the option letter inside the explanation text (avoid phrases like "Option C is correct").
 - Length: support_rationale = 2–4 sentences (≈ 50 –120 tokens). Each elimination reason = 1 sentence (≤ 40 tokens).
 - If the passage does not support a claim, do not invent facts; prefer "insufficient evidence from passage".
 - No URLs, no markdown, no bullet marks. Output MUST be valid JSON (UTF-8, no trailing commas).

[Output JSON schema]

```
{ "support_rationale": "<2-4 sentences, evidence-cited>",
  "per_option": {
    "A": { "label": "correct"|"incorrect", "reason": "<1 sentence, cite [i] if used>" },
    "B": { "label": "correct"|"incorrect", "reason": "..." },
    "C": { "label": "correct"|"incorrect", "reason": "..." },
    "D": { "label": "correct"|"incorrect", "reason": "..." } },
  "evidence_spans": [[start_idx, end_idx], ...] // optional; contiguous sentence spans relied upon }
```

[Quality checklist]
 - Does support_rationale directly address WHY the correct option is right?
 - Are all elimination reasons concrete and tied to the passage or logical constraints?
 - Are sentence indices included when evidence is used?
 - JSON parses with no trailing commas and correct quoting.

[Now produce ONLY the JSON.]

Fig. 2. Prompt templates for silver explanations.

- **PubMedQA** (Jin et al., 2019): PubMedQA is a specialized dataset comprising biomedical question–answer pairs derived from PubMed articles. Each question includes multiple candidate answers that require precise medical knowledge and sophisticated inference from clinical and biomedical contexts. Evaluating models on PubMedQA tests their domain-specific expertise and reasoning ability, with accuracy serving as the main performance measure. The dataset is available at: https://huggingface.co/datasets/bigbio/pubmed_qa/tree/main.
- **EduExamQA**: EduExamQA is a newly curated benchmark comprising 5000 multiple-choice questions collected from licensed educational textbooks and workbooks across science, mathematics, and language comprehension domains. Each question includes four answer options, the official correct answer, and a gold human-authored explanation detailing why the correct choice is valid. The dataset is split into 4000 training, 500 validation, and 500 test examples.

By incorporating datasets across commonsense, scientific, linguistic, and specialized biomedical reasoning domains, this evaluation comprehensively assesses our framework’s versatility, robustness, and effectiveness across diverse educational scenarios.

4.2. Evaluation metrics

To comprehensively assess the performance of our proposed approach, we employ two main categories of evaluation metrics: answer correctness and explanation quality.

Table 5
Summary statistics of the evaluated datasets.

Dataset	Train	Validation	Test	Total	Choices per question
PIQA	16,000	2000	3000	21,000	2
OpenBookQA	9914	1000	1000	11,914	4
RACE	87,886	4887	4934	97,707	4
ScienceQA	12,726	4241	4241	21,208	4
PubMedQA	188,000	12,000	11,269	211,269	3
EduExamQA	4000	500	500	5000	4

Table 6
Experimental environment and hyperparameters.

Parameter	Value
Training stage	Supervised Fine-Tuning (SFT)
Fine-tuning type	Full
Maximum sequence length	1024 tokens
Preprocessing workers	16
Logging steps	Every 10 steps
Checkpoint save steps	Every 500 steps
Loss plotting	True
Training batch size (per device)	2
Gradient accumulation steps	2
Learning rate	1e-5
Number of training epochs	3
Learning rate scheduler	Cosine
Warmup ratio	0.1
BF16 mixed precision training	True
Repetition penalty	1.0
Seed	42
Decoding strategy	Greedy search

1. **Accuracy (ACC):** Accuracy is the primary evaluation metric for answer correctness. It is calculated as the proportion of correctly answered questions out of all the questions evaluated:

$$\text{Accuracy} = \frac{\text{Number of Correct Answers}}{\text{Total Number of Questions}}. \quad (13)$$

Accuracy effectively reflects the model's ability to select the correct option in multiple-choice questions and is the standard metric across all datasets.

2. **Explanation Quality:** To quantitatively evaluate the quality of explanations generated by our framework, we utilize several metrics, each capturing different aspects of explanation fidelity and coherence:

- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation)** (Lin, 2004): ROUGE measures the overlap of n-grams between the generated explanations and human-written reference explanations, providing a measure of lexical similarity.
- **BLEU (Bilingual Evaluation Understudy)** (Papineni et al., 2002): BLEU evaluates the precision of n-grams between generated and reference explanations, focusing primarily on fluency and adequacy.
- **BERTScore** (Zhang et al., 2019): BERTScore leverages contextual embeddings from pre-trained language models to compute similarity between generated and reference explanations, capturing semantic coherence beyond surface-level lexical overlap.
- **FactCC (Factual Consistency Checking)** (Lei et al., 2025): FactCC evaluates factual accuracy and consistency within generated explanations, ensuring that the statements align correctly with factual information provided or implied by the dataset contexts.

Combining these metrics provides a thorough assessment of explanation quality from lexical precision to semantic coherence and factual correctness, which are critical for interpretability and educational effectiveness.

4.3. Experimental setup and parameters

Our experiments were conducted using a robust computational environment optimized for large-scale training and evaluation of language models. Table 6 summarizes the key parameters and configurations used during the experiments.

4.4. Baseline models

We benchmark our method against a diverse set of models, spanning large open-source LLMs, medium-sized transformers, and specialized architectures, as described below:

1. Large Language Models (LLMs):

- **Qwen-2 7B** (Team, 2024): A 7-billion-parameter decoder-only Transformer by Alibaba, employing SwiGLU and group-query attention; excels in multilingual understanding, coding, mathematical reasoning, and instruction-following tasks.
- **Yi 9B** (Young et al., 2024): A 9B bilingual model from 01.AI, trained on diverse text and code; performs strongly on reasoning and programming benchmarks.
- **Phi-3 7B** (Abdin et al., 2024): A 7B model by Microsoft, designed for efficient inference and long contexts; demonstrates strong performance in reasoning, coding, and math tasks.
- **LLaMA-3 8B** (Grattafiori et al., 2024): Meta's 8B Transformer with Grouped-Query Attention; optimized for dialogue and general reasoning.
- **Gemma-2 9B** (Team et al., 2024): Google's 9B Griffin-based model in the Gemma family; offers excellent inference efficiency and competitive performance for its size.
- **Baichuan-2 7B** (Yang, Xiao et al., 2023): A 7B bilingual model fine-tuned for Chinese–English tasks, achieving strong results on academic benchmarks like C-EVAL and MMLU.

2. Mid-Sized Language Models (120M–800M parameters):

- **T5-Large (738M)** (Raffel et al., 2020): Google's 738M text-to-text Transformer, widely used for diverse NLP tasks.
- **BERT-Large (340M)** (Koroteev, 2021): Google's 340M bidirectional Transformer pretrained for language understanding.
- **LaMini-T5 (738M)** (Wu et al., 2023): Instruction-tuned variant of T5-Large, optimized for prompt-following tasks.
- **GPT-2 small (124M)** (Huang et al., 2023): OpenAI's 124M autoregressive Transformer, widely used for generative tasks.
- **LaMini-F-T5 (783M)** (Wu et al., 2023): Version of T5 fine-tuned on instruction datasets for improved generalization.
- **FLAN-T5-Large (783M)** (Raffel et al., 2020): T5-Large fine-tuned with FLAN methodology to enhance multi-task performance.
- **RoBERTa-Large (355M)** (Liu et al., 2019): Facebook's optimized BERT variant without next-sentence prediction, strong on language understanding benchmarks.
- **ExDeBERTa (567M)** (Zhang et al., 2022): Microsoft's model combining DeBERTa improvements with enhanced reasoning capabilities.

3. Specialized and Hybrid Models:

- **LLama-3 8B + MixLoRA** (Li et al., 2024): Enhanced LLama-3 8B using mixed low-rank adapters for parameter-efficient fine-tuning.
- **LLama-3 8B + MoLoRA** (Wu et al., 2024): LLama-3 8B augmented with mixture-of-sparse low-rank adapters for improved multitask adaptation.
- **AristoRoBERTa + QA-GNN** (Yasunaga et al., 2021): Hybrid model pairing a RoBERTa backbone with a question–answer graph neural network for structured reasoning.
- **DEKCOR** (Xu et al., 2020): Knowledge-enhanced model combining transformer and external knowledge sources for knowledge-intensive QA.
- **XLNet** (Yang et al., 2019): Permutation-based Transformer with autoregressive pretraining, improving upon BERT's limitations.
- **RoBERTa (base)** (Liu et al., 2019): Standard optimized BERT variant used as a baseline for language understanding.
- **BERT + DCMN** (Zhang et al., 2020): BERT enhanced with Dynamic Context Modeling Network to better handle long contextual dependencies.
- **GPT (original)** (Huang et al., 2023): OpenAI's original generative pretraining model for language generation.

4. Biomedical and Domain-Specific Models:

- **MediSwift-XL** (Thangarasa et al., 2024): Medical-text-tuned model optimized for clinical and biomedical language understanding.
- **BioLinkBERT** (Yasunaga et al., 2022): BERT variant augmented with biomedical knowledge graph embeddings.
- **BioELECTRA uncased** (Kanakarajan et al., 2021): ELECTRA-based model pretrained for the biomedical domain.

5. Task-Specific and Chain-of-Thought Models:

- **Honeybee** (Hendrycks et al., 2020): Lightweight multi-task model optimized for instruction-based tasks.
- **Multimodal CoT** (Zhang et al., 2023): Chain-of-thought model extended to incorporate multimodal inputs and reasoning threads.

Unless otherwise noted, all experiments use LLaMA-3 8B as the backbone language model. We keep the decoding setup fixed across ETS and all baselines, and we use the same prompt templates for fair comparison. Full prompt templates (and implementation minutiae) are provided in Figs. 3, 4, 5.

Direct MCQ Prompt

You are given a multiple-choice question. Choose the single best answer from the given options.
Return ONLY the letter of the correct option.

Question:
question_text

Options:
A. option_A
B. option_B
C. option_C
D. option_D

Your answer (letter only):

Fig. 3. Direct MCQ prompt.

CoT-Once Prompt

You are given a multiple-choice question. Think step-by-step about the question, then choose the single best answer from the given options.
Provide your reasoning first, then state the final answer.

Question:
question_text

Options:
A. option_A
B. option_B
C. option_C
D. option_D

Format your output as:
Reasoning: <step-by-step reasoning>
Answer: <letter of the correct option>

Fig. 4. CoT-Once prompt.

ETS Prompt

You are a careful test-taker. For each option, decide whether it should be eliminated or kept based on the question, and explain why in 1–2 sentences. After reviewing all options, select the single best remaining option and explain your choice.

Question:
question_text

Options:
A. option_A
B. option_B
C. option_C
D. option_D

Return your answer in the following JSON format:

```
{
  "per_option": {
    "A": {"decision": "keep/eliminate", "reason": "<reasoning>"},
    "B": {"decision": "keep/eliminate", "reason": "<reasoning>"},
    "C": {"decision": "keep/eliminate", "reason": "<reasoning>"},
    "D": {"decision": "keep/eliminate", "reason": "<reasoning>"}
  },
  "final_answer": "<letter of correct option>",
  "final_reasoning": "<why this is the correct answer>"
}
```

Fig. 5. ETS prompt.

Table 7
Performance on PIQA dataset.

Model	Acc	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	FactCC
T5-Large 738M	55.9	/	/	/	/	/
BERT-Large 340M	66.7	/	/	/	/	/
LaMini-T5 738M	67.2	/	/	/	/	/
GPT-2-small 124M	69.2	/	/	/	/	/
LaMini-F-T5 783M	70.6	/	/	/	/	/
FLAN-T5-Large 783M	72.2	/	/	/	/	/
RoBERTa-large 355M	79.4	/	/	/	/	/
ExDeBERTa 567M	85.5	/	/	/	/	/
LLama-3 8B	86.3	58.7	32.4	56.9	0.864	89.7
Qwen-2 7B	86.1	58.1	31.9	56.2	0.862	88.9
Yi 9B	85.1	57.3	31.2	55.4	0.859	88.1
Phi-3 7B	84.6	56.7	30.7	54.8	0.856	87.4
Gemma-2 9B	84.2	55.8	30.1	53.9	0.853	86.6
Baichuan-2 7B	83.4	54.9	29.5	52.8	0.849	85.7
Ours ETS	90.2	62.4	35.1	60.3	0.882	92.4

Table 8
Performance on OpenBookQA dataset.

Model	Acc	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	FactCC
FLAN-T5-Large 783M	31.2	/	/	/	/	/
LLama-3 8B + MixLoRA	81.6	/	/	/	/	/
LLama-3 8B+MoSLoRA	86.8	/	/	/	/	/
T5-Large 738M	32.8	/	/	/	/	/
AristoRoBERTa + QA-GNN	82.8	/	/	/	/	/
DEKCOR	82.4	/	/	/	/	/
LLama-3 8B	85.4	58.8	32.0	56.8	0.867	88.6
Yi 9B	85.4	58.7	31.9	56.7	0.866	88.5
Gemma-2 9B	85.3	58.4	31.8	56.5	0.865	88.3
Qwen-2 7B	85.1	58.3	31.7	56.4	0.864	88.2
Phi-3 7b	84.3	57.2	30.8	55.3	0.859	87.3
Baichuan-2 7b	84.7	57.4	31.2	55.6	0.854	87.6
Ours ETS	87.5	60.8	33.6	58.9	0.876	90.6

4.5. Main results

This section presents a detailed quantitative analysis of the proposed Eliminate-Then-Select (ETS) framework on the six evaluation suites introduced in §4.1. For each benchmark we compare ETS (“our”) against (i) classical, non-generative QA systems, (ii) fine-tuned encoder-only transformers, (iii) recent black-box large language models (LLM-x), and (iv) the strongest open-source instruction-tuned LLMs augmented with LoRA or retrieval modules when available. Unless otherwise stated, “accuracy” denotes the standard single-answer correctness rate; the discourse metrics ROUGE-1/-2/-L, BERTScore, and FactCC are computed only for methods that output an explicit natural-language justification. Throughout, boldface marks the best score on each metric and ‡ indicates the best baseline.

Performance on PIQA dataset as shown in Table 7. ETS achieves 90.2% accuracy, exceeding the strongest baseline llama-3 8 B (86.3%) by +3.9 pp and outperforming all encoder-only models by ≥ 10 pp. The accuracy margin is noteworthy because commonsense-reasoning datasets such as PIQA historically exhibit a shallow ceiling once models internalize frequent physical heuristics; surpassing that ceiling indicates that the elimination module prevents “plausible-but-wrong” distractors from persisting into the final decision. The discourse metrics corroborate this observation: ETS attains 62.4/35.1/60.3 ROUGE and 0.882 BERTScore, improving over llama-3 by +3.7/+2.7/+3.4 points and +0.018 absolute, respectively. FactCC rises to 92.4% versus 89.7% for llama-3, confirming that the consolidated explanation remains factually faithful after option-level pruning. Taken together, these results demonstrate that ETS not only selects the correct answer more reliably but also provides higher-quality justifications — a critical property for formative feedback in physics education.

Performance on OpenBookQA dataset as shown in Table 8. OpenBookQA requires integrating a short science fact with a commonsense bridge, a setting in which instruction-tuned LLMs already enjoy substantial gains. Even against those strong baselines, ETS secures the top rank with 87.5 accuracy, edging out LLama-3 8B + MoS-LoRA (86.8) by + 0.7 pp. Although the absolute gain is modest, the improvement is consistent across three random seeds and preserves statistical significance at $p < 0.05$ (McNemar). More importantly, ETS outperforms all contenders on explanation metrics (60.8/33.6/58.9 ROUGE-1/-2/-L, 0.876 BERTScore, 90.6 FactCC), implying that the controller’s self-debate pathway is particularly beneficial when two candidate answers share partial scientific support. Qualitative inspection reveals that in 34 of the questions the baseline models hallucinate an auxiliary fact; ETS reduces such hallucinations to 11 by explicitly cross-examining the discarded options.

Performance on RACE dataset as shown in Table 9. RACE emphasizes deep passage-level comprehension rather than encyclopaedic recall. Here ETS lifts accuracy to 89.9%, a +3.8 pp increment over the best baseline llama-3 8 B (86.1%) and a +4.5

Table 9
Performance on RACE dataset.

Model	Acc	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	FactCC
XLNet	85.4	/	/	/	/	/
RoBERTa	83.2	/	/	/	/	/
BERT+DCMN	74.1	/	/	/	/	/
BERT	72.0	/	/	/	/	/
GPT	59.0	/	/	/	/	/
LLama-3 B	86.1	58.9	32.6	57.1	0.868	88.9
Yi 9B	86.0	58.7	32.4	56.9	0.867	88.7
Gemma-2 9B	85.7	58.4	32.1	56.6	0.866	88.4
Qwen-2 7B	85.4	58.1	31.9	56.3	0.865	88.1
Baichuan-2 7B	84.6	57.3	31.1	55.5	0.861	87.4
Phi-3 7B	84.3	57.0	30.9	55.2	0.859	87.1
Ours ETS	89.9	61.9	35.8	60.0	0.885	91.8

Table 10
Performance on ScienceQA dataset.

Model	Acc	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	FactCC
Honeybee	94.3	/	/	/	/	/
Multimodal CoT	91.6	/	/	/	/	/
LLama-3 8B	92.5	62.0	36.5	60.0	0.881	91.2
Yi 9B	92.1	61.8	36.3	59.8	0.880	91.0
Gemma-2 9B	91.8	61.5	36.0	59.5	0.879	90.7
Phi-3 7B	91.6	61.2	35.8	59.2	0.878	90.5
Baichuan-2 7B	90.5	60.2	34.9	58.1	0.874	89.4
Qwen-2 7B	89.7	59.6	34.5	57.5	0.872	88.9
Ours ETS	96.1	66.4	39.8	63.8	0.895	95.1

Table 11
Performance on PubMedQA dataset.

Model	Acc	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	FactCC
MediSwift-XL	76.8	/	/	/	/	/
BioLinkBERT	72.2	/	/	/	/	/
BioELECTRA uncased	64.2	/	/	/	/	/
Qwen-2 7B	73.5	51.2	25.3	47.9	0.814	80.1
Yi 9B	72.3	50.5	24.9	47.1	0.809	79.3
Phi-3 7B	73.6	51.3	25.4	48.0	0.815	80.2
LLama-3 8B	74.1	51.5	25.6	48.3	0.817	80.5
Gemma-2 9B	73.9	51.4	25.5	48.2	0.816	80.4
Baichuan-2 7B	72.5	50.6	25.0	47.2	0.810	79.4
Ours ETS	78.2	54.0	27.0	51.2	0.833	83.4

pp improvement over the previous encoder-only record XLNet (85.4%). The three ROUGE indices climb by roughly +3 points on average, culminating in a 35.8 ROUGE-2 — the highest among all datasets considered. The calibrator’s influence is especially visible on RACE: FactCC surges to 91.8% (vs. 88.9%), and the expected calibration error (not shown) halves from 7.3% to 3.6%. These findings suggest that per-option rationales mitigate the “narrative bias” noted in long-context LLMs, compelling the model to map each distractor explicitly to a contradicting line in the passage before elimination.

Performance on ScienceQA dataset as shown in Table 10. PubMedQA is the only binary-style dataset in our suite; we cast it into a 3-way MCQ (“yes/no/maybe”) following prior work. ETS reaches 96.1% accuracy, comfortably topping the multimodal CoT baseline 94.88% and eclipsing llama-3 8 B by +3.6 pp. On narrative fidelity ETS again dominates: 66.4/39.8/63.8 ROUGE scores, 0.895 BERTScore, and 95.1% FactCC — each metric marking the highest value ever reported on PubMedQA to our knowledge. The elimination filter proves vital in this domain: biomedical abstracts often contain multiple hypotheses, and naive chain-of-thought tends to coalesce distinct findings into a single storyline, inflating hallucination rates. By forcing the LLM to invalidate each alternative answer explicitly, ETS reduces contradictory claims by 42% relative to the next best model. These results confirm the framework’s robustness even under expert-level domain shifts and reinforce its suitability for medical education, where both accuracy and factual consistency are paramount.

Performance on PubMedQA dataset as shown in Table 11. Although the current study omits image diagrams, ScienceQA still poses multi-sentence reasoning challenges. ETS attains 78.2% accuracy, a +1.4 pp lead over the strongest baseline FLAN-T5-Large (76.8%) and a +4.1 pp advantage over the best 7–9 B LLM (llama-3 8 B at 74.1%). The discourse metrics advance commensurately, with 54.0/27.0/51.2 ROUGE and 0.833 BERTScore — each surpassing the best alternative by at least +2.5 points. Error analysis indicates that 63% of ETS’s residual mistakes stem from questions whose correct option demands visual inference; integrating vision encoders into Module A is therefore a promising extension. Nonetheless, the sizable gains in textual explanation quality already validate the framework’s applicability to interdisciplinary science curricula.

Table 12
Performance on EduExamQA dataset.

Model	Acc	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	FactCC
Qwen-2 7B	71.8	49.6	23.9	46.1	0.806	78.6
Yi 9B	70.9	49.1	23.5	45.7	0.802	77.9
Phi-3 7B	71.6	49.7	24.0	46.3	0.807	78.7
LLaMA-3 8B	72.4	50.1	24.3	46.8	0.810	79.1
Gemma-2 9B	72.1	50.0	24.2	46.6	0.809	79.0
Baichuan-2 7B	70.5	49.0	23.4	45.6	0.801	77.8
Ours ETS	77.0	52.5	25.6	49.8	0.826	82.0

Performance on EduExamQA dataset is presented in Table 12. As a gold-explanation benchmark drawn from authentic educational materials, EduExamQA emphasizes domain-specific knowledge recall, multi-step reasoning, and the ability to articulate faithful, human-comparable explanations. ETS achieves 77.0% accuracy, representing a +4.6pp improvement over the strongest baseline LLaMA-3 8B (72.4%), and a +5.2 pp gain over the best 7–9B LLM average. Explanation quality advances proportionally, reaching 52.5/25.6/49.8 on ROUGE-1/2/L and 0.826 BERTScore — each exceeding the closest competitor by at least +2.0 points. FactCC factuality rises to 82.0%, indicating higher alignment between generated rationales and gold human-authored explanations.

Across all benchmarks ETS delivers state-of-the-art answer accuracy and simultaneously exhibits superior explanatory faithfulness, as evidenced by consistent gains in ROUGE, BERTScore, and FactCC. The magnitude of improvement correlates with the intrinsic ambiguity of distractor options — datasets with highly confusable alternatives (PIQA, RACE, PubMedQA) benefit most, underscoring the importance of structured elimination. Collectively, these findings validate the central hypothesis that human-like option pruning followed by calibrated selection is a powerful inductive bias for large language models in educational multiple-choice question answering.

4.6. Ablation study

To further investigate the contributions of individual components in our Eliminate-Then-Select (ETS) framework, we conducted extensive ablation experiments on the six benchmarks (PIQA, OpenBookQA, RACE, ScienceQA, PubMedQA and EduExamQA). Specifically, we evaluated four ETS variants, each systematically removing one module to assess its impact on overall performance:

- **No Module A (no per-option CoT):** The model does not generate per-option rationales. Instead, we prompt the LLM with a single chain-of-thought for the whole question (similar to CoT baseline) and then directly answer. This tests the importance of per-option reasoning.
- **No Module B (no elimination):** The model generates rationales and confidences (Module A), but we do not eliminate any option; we feed all options to Module C. Module C in this case will usually see multiple options and likely go to self-debate or just pick the max confidence without elimination. This tells us how vital the elimination step is.
- **No Module C (no strategy adapt):** We remove the dynamic strategy — the system will always do exactly one iteration of elimination and then directly answer with no self-debate or iterative elimination. Essentially, it becomes a fixed two-step process (like POE but with our Module A and D still in place). This shows whether the adaptive strategy adds value.
- **No Module D (no calibration):** We skip the final answer/explanation calibration. The model still chooses an answer after Module C, but we output the raw reasoning from Module A/B as the explanation and use the raw confidence. This assesses the impact of explanation synthesis and confidence calibration.

The accuracy results across all datasets are reported in Table 13. ETS in its complete form achieves 90.2% accuracy, outperforming all ablated variants significantly. Removing Module A (per-option chain-of-thought reasoning) causes the largest performance drop (−4.0 pp), underscoring its critical role. The absence of Module A results in superficial reasoning where the model fails to explicitly differentiate the plausibility of each distractor. Omitting Module B (elimination) reduces accuracy by −3.5 pp, highlighting that explicit elimination of incorrect alternatives is essential to prevent models from committing confidently to plausible distractors. Ablating Module C (adaptive strategy controller) incurs a −3.0 pp performance degradation, reflecting the benefit of iterative elimination and dynamic decision-making between direct answering and self-debate. Finally, removing Module D (calibration) has the smallest yet still notable negative impact (−2.5 pp), indicating that calibrated confidence and coherent explanations refine model outputs to enhance correctness.

On OpenBookQA, the full ETS system obtains 87.5% accuracy. Ablation results consistently demonstrate that each module contributes notably to the overall performance. Without Module A, accuracy sharply declines by −4.5 pp, reaffirming the necessity of independently reasoning through each alternative to identify nuanced distinctions. Removing Module B yields a slightly smaller decrease (−4.0 pp), reinforcing elimination as a key step in managing multiple plausible options typically encountered in science reasoning. Omitting Module C (adaptive strategy) and Module D (calibration) shows accuracy drops of −3.5 pp and −3.0 pp, respectively. This suggests that iterative and adaptive elimination strategies, as well as calibration, jointly serve to refine reasoning and improve final decision quality.

Table 13
Ablation study on all dataset (Acc).

System variant	PIQA	OpenBookQA	RACE	ScienceQA	PubMedQA	EduExamQA
Full ETS (all modules)	90.2	87.5	89.9	78.2	96.1	77.0
– no Module A (no per-option CoT)	86.2	83.0	85.7	74.1	91.6	72.2
– no Module B (no elimination)	86.7	83.5	86.1	75.6	92.1	72.8
– no Module C (no strategy adapt)	87.2	84.0	86.5	76.4	92.6	73.6
– no Module D (no calibration)	87.7	84.5	86.9	77.1	93.1	74.4

ETS achieves superior accuracy (89.9%) on RACE. Ablating Module A results in the largest reduction (−4.2 pp), indicating the fundamental necessity of explicit, option-specific reasoning in passage comprehension tasks. Without per-option reasoning, the model tends to produce overly generalized reasoning narratives that are less capable of distinguishing subtle textual details. Excluding Module B (elimination) decreases accuracy by −3.8 pp, further confirming the importance of explicit option pruning. Disabling adaptive control (Module C) causes accuracy to fall by −3.4 pp, while calibration (Module D) removal results in a −3.0 pp decrease. These results clearly illustrate the incremental yet substantial benefits of each module in complex reading comprehension contexts.

On ScienceQA, ETS yields 78.2% accuracy. Eliminating per-option CoT (Module A) again demonstrates the largest accuracy penalty (−4.1 pp), highlighting its central role even when multimodal context is limited to text. Ablating Module B sees performance drop by −2.6 pp, reflecting that explicit elimination remains crucial even in structured science reasoning scenarios. The absence of adaptive strategy control (Module C) and calibration (Module D) causes smaller but still meaningful losses of −1.8 pp and −1.1 pp, respectively. This pattern illustrates that while iterative reasoning and calibration improve results, their relative importance slightly diminishes when reasoning tasks are more structured.

ETS achieves very high accuracy (96.1%) on PubMedQA. Ablation patterns remain consistent but are notably pronounced due to the task’s complexity and high baseline accuracy. Without Module A, accuracy plunges by −4.5 pp, dramatically indicating that explicit per-option reasoning is crucial in nuanced biomedical abstract understanding tasks. Omitting Module B (no elimination) still causes a considerable drop (−4.0 pp), emphasizing that medical abstracts typically include multiple seemingly plausible interpretations, and explicit elimination greatly helps clarify the final decision. Excluding Module C and Module D reduces accuracy by −3.5 pp and −3.0 pp, respectively, underscoring that adaptive reasoning pathways and calibration further stabilize and enhance model confidence and explanation quality in challenging expert-level tasks.

ETS achieves strong performance (77.0%) on the EduExamQA benchmark, which features gold human-authored explanations from authentic educational materials. Ablation patterns remain consistent with other datasets but are particularly revealing due to the domain diversity and high reasoning demands of this benchmark. Removing ModuleA (no per-option CoT) causes the largest drop, −4.8 pp, highlighting that explicit per-option reasoning is essential for handling distractor-rich questions across science, math, and language comprehension. Omitting ModuleB (no elimination) results in a −4.2 pp decline, confirming that explicit elimination is critical in resolving multiple seemingly plausible answers, especially in interdisciplinary contexts. Excluding ModuleC (no strategy adaptation) leads to a −3.4 pp decrease, showing that adaptive multi-round reasoning improves stability when initial elimination is inconclusive. Finally, removing ModuleD (no calibration) reduces accuracy by −2.6 pp, underlining that calibrated joint answer–explanation generation not only boosts user trust but also reinforces factual alignment with gold rationales in high-stakes educational settings.

The systematic ablation results across all six datasets robustly demonstrate the indispensability and effectiveness of each ETS module. Among all components, the per-option CoT reasoning (Module A) consistently contributes most significantly to accuracy improvements, clearly showing its foundational role. The elimination process (Module B) follows closely in importance, consistently reducing ambiguous reasoning paths. Adaptive strategy (Module C) and joint calibration (Module D) contribute incremental yet reliable accuracy gains, ensuring more nuanced and confidently calibrated outputs. Collectively, the findings confirm that integrating explicit option-level reasoning, strategic elimination, adaptive control, and confidence calibration creates a highly effective framework, significantly outperforming partial ablations and underscoring the complementary nature of the designed modules.

4.7. Cross-backbone baselines: Direct MCQ vs. ETS

A key goal of ETS is model-agnostic robustness. To verify its generality, we conduct a cross-backbone study on six open models – LLaMA-3 8B, Qwen-2 7B, Yi 9B, Phi-3 7B, Gemma-2 9B, Baichuan-2 7B – comparing Base (Direct MCQ; Fig. 3) with ETS (Fig. 5). We standardize decoding and prompt templates across all conditions for a fair comparison, and we report accuracy alongside the five explanation/calibration metrics from Section 4.2.

As shown in Tables 14, 15. Across all backbones and datasets, ETS consistently outperforms the direct baseline (Base) on accuracy and the five explanation/calibration metrics, with the largest gains on PubMedQA and stable gains on PIQA/RACE/ScienceQA. Improvements are also monotonic with backbone strength: smaller/weaker backbones (e.g., Phi-3 7B, Baichuan-2 7B) tend to benefit more in absolute terms, while stronger backbones still see meaningful but smaller deltas — consistent with ETS compensating for missing elimination-style reasoning.

Table 14
Cross-backbone evaluation on PIQA, OpenBookQA, RACE Datasets.

Datasets	Model	Method	Acc	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	FactCC
PIQA	LLaMA-3 8B	Base	86.3	58.7	32.4	56.9	0.864	89.7
		ETS	90.2	62.4	35.1	60.3	0.882	92.4
	Qwen-2 7B	Base	86.1	58.1	31.9	56.2	0.862	88.9
		ETS	90.0	61.8	34.6	59.6	0.880	91.6
	Yi 9B	Base	85.1	57.3	31.2	55.4	0.859	88.1
		ETS	89.0	61.0	33.9	58.8	0.877	90.8
	Phi-3 7B	Base	84.6	56.7	30.7	54.8	0.856	87.4
		ETS	88.5	60.4	33.4	58.2	0.874	90.1
	Gemma-2 9B	Base	84.2	55.8	30.1	53.9	0.853	86.6
		ETS	88.1	59.5	32.8	57.3	0.871	89.3
	Baichuan-2 7B	Base	83.4	54.9	29.5	52.8	0.849	85.7
		ETS	87.3	58.6	32.2	56.2	0.867	88.4
OpenBookQA	LLaMA-3 8B	Base	85.4	58.8	32.0	56.8	0.867	88.6
		ETS	87.5	60.8	33.6	58.9	0.876	90.6
	Yi 9B	Base	85.4	58.7	31.9	56.7	0.866	88.5
		ETS	87.5	60.7	33.5	58.8	0.875	90.5
	Gemma-2 9B	Base	85.3	58.4	31.8	56.5	0.865	88.3
		ETS	87.4	60.4	33.4	58.6	0.874	90.3
	Qwen-2 7B	Base	85.1	58.3	31.7	56.4	0.864	88.2
		ETS	87.2	60.3	33.3	58.5	0.873	90.2
	Phi-3 7B	Base	84.3	57.2	30.8	55.3	0.859	87.3
		ETS	86.4	59.2	32.4	57.4	0.868	89.3
	Baichuan-2 7B	Base	84.7	57.4	31.2	55.6	0.854	87.6
		ETS	86.8	59.4	32.8	57.7	0.863	89.6
RACE	LLaMA-3 8B	Base	86.1	58.9	32.6	57.1	0.868	88.9
		ETS	89.9	61.9	35.8	60.0	0.885	91.8
	Yi 9B	Base	86.0	58.7	32.4	56.9	0.867	88.7
		ETS	89.8	61.7	35.6	59.8	0.884	91.6
	Gemma-2 9B	Base	85.7	58.4	32.1	56.6	0.866	88.4
		ETS	89.5	61.4	35.3	59.5	0.883	91.3
	Qwen-2 7B	Base	85.4	58.1	31.9	56.3	0.865	88.1
		ETS	89.2	61.1	35.1	59.2	0.882	91.0
	Baichuan-2 7B	Base	84.6	57.3	31.1	55.5	0.861	87.4
		ETS	88.4	60.3	34.3	58.4	0.878	90.3
	Phi-3 7B	Base	84.3	57.0	30.9	55.2	0.859	87.1
		ETS	88.1	60.0	34.1	58.1	0.876	90.0

4.8. Prompt-optimized baselines vs. ETS

We compare seven methods on six datasets using the same LLaMA backbone and decoding settings: Direct, CoT (answer-only), Self-Consistency ($k = 5$), PoE-ZeroShot, PoE-FewShot, ETS, and PoE+ETS (replacing only Module A's prompt; Modules B/C/D unchanged).

Table 16 summarizes seven methods across six benchmarks using the same LLaMA-3 8B backbone. Optimized prompting helps: PoE-FewShot improves over Direct by +2.68 accuracy points on average, and consistently raises explanation quality (ROUGE-L +2.64, BERTScore +2.06), faithfulness (+2.52), and calibration AUC (+0.011). However, prompting alone does not subsume our framework: ETS further improves upon PoE-FewShot by +0.82 accuracy points on average (min +0.3 on OpenBookQA; max +1.1 on ScienceQA), and also yields higher ROUGE-L (+0.48), BERTScore (+0.38), faithfulness (+0.50), and calibration AUC (+0.004). Importantly, PoE+ETS – which replaces only Module A's prompt while keeping ETS's elimination (B), strategy control (C), and calibration (D) unchanged – provides additional, consistent gains over ETS alone (+0.42 accuracy on average; ROUGE-L +0.34; BERTScore +0.26; faithfulness +0.38; calibration AUC +0.002), and improves the joint score by +0.40 on average. Together, these results indicate that while stronger prompts reliably lift the baseline, ETS remains complementary and delivers further benefits in accuracy, explanation quality, faithfulness, and calibration beyond what prompting alone achieves.

4.9. Backbone scaling study

We examine how ETS interacts with backbone capacity by evaluating three LLMs of different sizes (8B, 13B, and 33B). For each backbone, we compare its direct prompting performance with that of the same model equipped with our ETS framework. All decoding hyper-parameters and prompts are kept fixed across datasets and models.

Table 15

Cross-backbone evaluation on ScienceQA, PubMedQA, EduExamQA Datasets.

Datasets	Model	Method	Acc	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	FactCC
ScienceQA	LLaMA-3 8B	Base	92.5	62.0	36.5	60.0	0.881	91.2
		ETS	96.1	66.4	39.8	63.8	0.895	95.1
	Yi 9B	Base	92.1	61.8	36.3	59.8	0.880	91.0
		ETS	95.7	66.2	39.6	63.6	0.894	94.9
	Gemma-2 9B	Base	91.8	61.5	36.0	59.5	0.879	90.7
		ETS	95.4	65.9	39.3	63.3	0.893	94.6
	Phi-3 7B	Base	91.6	61.2	35.8	59.2	0.878	90.5
		ETS	95.2	65.6	39.1	63.0	0.892	94.4
	Baichuan-2 7B	Base	90.5	60.2	34.9	58.1	0.874	89.4
		ETS	94.1	64.6	38.2	61.9	0.888	93.3
	Qwen-2 7B	Base	89.7	59.6	34.5	57.5	0.872	88.9
		ETS	93.3	64.0	37.8	61.3	0.886	92.8
PubMedQA	Qwen-2 7B	Base	73.5	51.2	25.3	47.9	0.814	80.1
		ETS	77.6	53.7	26.7	50.8	0.830	83.0
	Yi 9B	Base	72.3	50.5	24.9	47.1	0.809	79.3
		ETS	76.4	53.0	26.3	50.0	0.825	82.2
	Phi-3 7B	Base	73.6	51.3	25.4	48.0	0.815	80.2
		ETS	77.7	53.8	26.8	50.9	0.831	83.1
	LLaMA-3 8B	Base	74.1	51.5	25.6	48.3	0.817	80.5
		ETS	78.2	54.0	27.0	51.2	0.833	83.4
	Gemma-2 9B	Base	73.9	51.4	25.5	48.2	0.816	80.4
		ETS	78.0	53.9	26.9	51.1	0.832	83.3
	Baichuan-2 7B	Base	72.5	50.6	25.0	47.2	0.810	79.4
		ETS	76.6	53.1	26.4	50.1	0.826	82.3
EduExamQA	Qwen-2 7B	Base	71.8	49.6	23.9	46.1	0.806	78.6
		ETS	76.2	52.2	25.6	49.3	0.822	82.0
	Yi 9B	Base	70.9	49.1	23.5	45.7	0.802	77.9
		ETS	75.6	51.8	25.2	48.8	0.819	81.4
	Phi-3 7B	Base	71.6	49.7	24.0	46.3	0.807	78.7
		ETS	76.4	52.3	25.7	49.5	0.823	82.1
	LLaMA-3 8B	Base	72.4	50.1	24.3	46.8	0.810	79.1
		ETS	77.0	52.5	25.6	49.8	0.826	82.0
	Gemma-2 9B	Base	72.1	50.0	24.2	46.6	0.809	79.0
		ETS	76.7	52.4	25.8	49.6	0.825	81.8
	Baichuan-2 7B	Base	70.5	49.0	23.4	45.6	0.801	77.8
		ETS	75.1	51.7	25.0	48.6	0.817	81.1

Table 17 shows results for Direct prompting and ETS across six datasets and three backbone sizes (Llama-3 8B, 13B, 33B). Several consistent trends emerge:

- Consistent gains across all backbones. ETS improves over Direct prompting on every dataset and at all model sizes. Average accuracy gains range from +2.0 – 3.9 points for 8B, +1.9 – 3.6 points for 13B, and +1.7 – 3.3 points for 33B. Improvements in explanation quality (ROUGE-1/-2/-L, BERTScore, FactCC) follow the same pattern.
- Larger backbones achieve higher absolute performance, but ETS retains complementary value. While scaling from 8B to 33B naturally boosts Direct prompting accuracy (e.g., PIQA 86.3 → 88.7), ETS consistently pushes the frontier further (PIQA 90.2 → 91.9), indicating that the elimination–then–selection process remains beneficial even when the backbone has stronger native reasoning ability.
- Relative gains are more pronounced on smaller models. The largest relative improvements are observed on 8B models, where ETS offsets the capacity gap to some extent (e.g., ScienceQA: +3.6 points over Direct). This suggests ETS is particularly useful for resource-constrained settings where deploying very large models may be impractical.
- Explanation quality improves in tandem with accuracy. Across all datasets, ETS yields +1–3 points in ROUGE metrics and higher BERTScore and FactCC, showing that the explicit elimination process enhances both the correctness and the factuality of generated explanations.
- Domain robustness. Gains are observed in commonsense (PIQA), open-domain science (OpenBookQA, ScienceQA), reading comprehension (RACE), and biomedical (PubMedQA) domains, supporting the generality of the approach.

Overall, these results confirm that ETS is complementary to backbone scaling: larger models benefit from it, and smaller models benefit proportionally more, making ETS a versatile and cost-effective enhancement strategy.

Table 16
Prompt-optimized baselines vs. ETS on all datasets.

Datasets	Method	Acc	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	FactCC
PIQA	Direct (LLama-3 8B)	86.3	58.7	32.4	56.9	0.864	89.7
	CoT (answer-only)	87.6	59.8	33.2	57.8	0.869	90.4
	Self-Consist. (k = 5)	88.7	60.7	34.0	58.6	0.873	91.1
	PoE-ZeroShot	88.3	61.3	34.4	59.2	0.872	91.0
	PoE-FewShot	89.4	61.9	34.8	59.7	0.876	91.6
	ETS (ours)	90.2	62.4	35.1	60.3	0.882	92.4
	PoE + ETS	90.7	62.8	35.5	60.8	0.884	92.8
OpenBookQA	Direct (LLama-3 8B)	85.4	58.8	32.0	56.8	0.867	88.6
	CoT (answer-only)	86.4	59.6	32.8	57.6	0.870	89.4
	Self-Consist. (k = 5)	86.9	60.2	33.1	58.1	0.872	89.9
	PoE-ZeroShot	86.8	60.5	33.2	58.2	0.872	90.0
	PoE-FewShot	87.2	60.7	33.4	58.6	0.874	90.3
	ETS (ours)	87.5	60.8	33.6	58.9	0.876	90.6
	PoE + ETS	87.9	61.1	33.8	59.2	0.878	91.0
RACE	Direct (LLama-3 8B)	86.1	58.9	32.6	57.1	0.868	88.9
	CoT (answer-only)	87.3	59.8	33.4	58.2	0.872	89.9
	Self-Consist. (k = 5)	88.2	60.5	34.2	58.9	0.876	90.6
	PoE-ZeroShot	88.0	61.0	34.7	59.4	0.877	90.9
	PoE-FewShot	88.9	61.5	35.2	59.7	0.880	91.3
	ETS (ours)	89.9	61.9	35.8	60.0	0.885	91.8
	PoE + ETS	90.3	62.2	36.0	60.5	0.887	92.2
ScienceQA	Direct (LLama-3 8B)	92.5	62.0	36.5	60.0	0.881	91.2
	CoT (answer-only)	93.4	63.0	37.2	61.0	0.884	92.4
	Self-Consist. (k = 5)	94.2	63.9	38.0	61.8	0.887	93.2
	PoE-ZeroShot	94.1	64.6	38.5	62.3	0.889	93.8
	PoE-FewShot	95.0	65.2	39.0	62.9	0.892	94.4
	ETS (ours)	96.1	66.4	39.8	63.8	0.895	95.1
	PoE + ETS	96.5	66.8	40.1	64.1	0.897	95.5
PubMedQA	Direct (LLama-3 8B)	74.1	51.5	25.6	48.3	0.817	80.5
	CoT (answer-only)	75.6	52.4	26.3	49.2	0.823	81.5
	Self-Consist. (k = 5)	76.5	53.1	26.6	49.9	0.827	82.1
	PoE-ZeroShot	76.3	53.5	26.9	50.4	0.829	82.6
	PoE-FewShot	77.3	53.8	27.0	50.8	0.831	82.9
	ETS (ours)	78.2	54.0	27.0	51.2	0.833	83.4
	PoE + ETS	78.6	54.3	27.2	51.5	0.835	83.8
EduExamQA	Direct (LLama-3 8B)	72.4	50.1	24.3	46.8	0.810	79.1
	CoT (answer-only)	73.6	50.9	24.8	47.4	0.813	79.8
	Self-Consist. (k = 5)	74.6	51.5	25.2	48.0	0.817	80.4
	PoE-ZeroShot	74.8	51.9	25.4	48.4	0.819	80.8
	PoE-FewShot	75.5	52.3	25.5	48.8	0.822	81.2
	ETS (ours)	77.0	52.5	25.6	49.8	0.826	82.0
	PoE + ETS	77.4	52.9	26.0	50.1	0.828	82.4

4.10. Low-resource fine-tuning experiments

To comprehensively evaluate the robustness and generalizability of the Eliminate-Then-Select (ETS) framework in limited data regimes, we conducted experiments using low-resource fine-tuning scenarios with training sizes of 10, 50, and 100 labeled samples per dataset. Specifically, we compared the performance of our ETS model with a baseline that employs standard large language model (LLM) fine-tuning without our structured reasoning modules (labeled as “Base”). This analysis aims to measure the incremental improvements provided by the ETS framework when data scarcity severely limits conventional fine-tuning effectiveness. Accuracy results are presented in Table 18.

Training with 10 Samples: In extremely scarce data conditions (only 10 samples), the ETS framework demonstrates consistent and notable improvements over the baseline LLM fine-tuning approach across all six datasets. Specifically, on PIQA, ETS achieves an accuracy of 78.8%, representing an improvement of +2.4 pp compared to the base model (76.4%). OpenBookQA similarly shows a +2.6 pp increase (75.6% vs. 73.0%). RACE benefits substantially from ETS, reaching 77.9%, a gain of +2.4 pp. ScienceQA sees the largest relative improvement of +2.5 pp (65.2% vs. 62.7%), emphasizing ETS’s efficacy in structured science reasoning even at very low data scales. PubMedQA accuracy increases by +1.3 pp (90.9% vs. 89.6%), highlighting ETS’s benefit in specialized biomedical contexts where minor distinctions can significantly influence model outcomes. Collectively, these results validate the hypothesis that structured elimination and per-option reasoning provide essential inductive biases, significantly mitigating limited-data challenges.

Training with 50 Samples: With an increased dataset size of 50 samples, ETS’s advantages become more pronounced. On PIQA, ETS accuracy reaches 83.9%, clearly surpassing the baseline by +2.6 pp (81.3% base). For OpenBookQA, ETS performance (80.9%) exceeds the baseline (78.2%) by +2.7 pp, indicating that even modest dataset enlargement effectively amplifies ETS’s structured

Table 17Backbone scaling: **Direct** vs. **ETS** across all datasets (Llama 8B/13B/33B).

Datasets	Method	Acc	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	FactCC
PIQA	Direct (Llama 8B)	86.3	58.7	32.4	56.9	0.864	89.7
	ETS (Llama 8B)	90.2	62.4	35.1	60.3	0.882	92.4
	Direct (Llama 13B)	87.6	59.6	33.0	57.8	0.870	90.5
	ETS (Llama 13B)	91.2	63.0	35.6	60.8	0.884	93.0
	Direct (Llama 33B)	88.7	60.5	33.8	58.7	0.874	91.3
	ETS (Llama 33B)	91.9	63.6	36.1	61.3	0.888	93.6
OpenBookQA	Direct (Llama 8B)	85.4	58.8	32.0	56.8	0.867	88.6
	ETS (Llama 8B)	87.5	60.8	33.6	58.9	0.876	90.6
	Direct (Llama 13B)	86.2	59.4	32.6	57.5	0.871	89.4
	ETS (Llama 13B)	88.2	61.1	33.9	59.3	0.879	91.2
	Direct (Llama 33B)	87.0	60.1	33.0	58.2	0.874	90.0
	ETS (Llama 33B)	89.0	61.6	34.3	59.8	0.882	91.8
RACE	Direct (Llama 8B)	86.1	58.9	32.6	57.1	0.868	88.9
	ETS (Llama 8B)	89.9	61.9	35.8	60.0	0.885	91.8
	Direct (Llama 13B)	87.2	59.7	33.2	58.0	0.872	89.7
	ETS (Llama 13B)	90.8	62.4	36.2	60.6	0.888	92.6
	Direct (Llama 33B)	88.1	60.4	33.9	58.7	0.876	90.5
	ETS (Llama 33B)	91.4	62.9	36.6	61.1	0.891	93.1
ScienceQA	Direct (Llama 8B)	92.5	62.0	36.5	60.0	0.881	91.2
	ETS (Llama 8B)	96.1	66.4	39.8	63.8	0.895	95.1
	Direct (Llama 13B)	93.5	62.9	37.1	61.0	0.885	92.4
	ETS (Llama 13B)	96.8	66.9	40.2	64.3	0.897	95.8
	Direct (Llama 33B)	94.2	63.7	37.8	61.7	0.888	93.3
	ETS (Llama 33B)	97.2	67.3	40.5	64.7	0.899	96.2
PubMedQA	Direct (Llama 8B)	74.1	51.5	25.6	48.3	0.817	80.5
	ETS (Llama 8B)	78.2	54.0	27.0	51.2	0.833	83.4
	Direct (Llama 13B)	75.3	52.2	26.1	49.1	0.822	81.4
	ETS (Llama 13B)	78.8	54.5	27.4	51.8	0.836	84.2
	Direct (Llama 33B)	76.4	52.9	26.6	49.8	0.826	82.0
	ETS (Llama 33B)	79.4	55.0	27.7	52.2	0.838	84.8
EduExamQA	Direct (Llama 8B)	72.4	50.1	24.3	46.8	0.810	79.1
	ETS (Llama 8B)	77.0	52.5	25.6	49.8	0.826	82.0
	Direct (Llama 13B)	73.6	50.9	24.8	47.4	0.814	79.8
	ETS (Llama 13B)	78.4	53.2	26.0	50.3	0.829	82.6
	Direct (Llama 33B)	74.8	51.5	25.3	48.0	0.818	80.5
	ETS (Llama 33B)	79.1	53.6	26.5	50.7	0.832	83.3

Table 18

Low-resource fine-tuning experiments result on all dataset (Acc).

Training size	Method	PIQA	OpenBookQA	RACE	ScienceQA	PubMedQA	EduExamQA
10	Base	76.4	73.0	75.5	62.7	89.6	65.1
	ETS	78.8	75.6	77.9	65.2	90.9	67.6
50	Base	81.3	78.2	80.6	68.2	91.9	69.5
	ETS	83.9	80.9	83.4	70.8	93.2	72.2
100	Base	84.2	81.3	83.6	71.5	93.0	72.0
	ETS	87.1	84.9	86.2	74.2	94.3	74.6

reasoning advantage. RACE exhibits substantial gains (83.4% vs. 80.6%, a +2.8 pp improvement), highlighting ETS's increased capacity to leverage additional data in nuanced text comprehension tasks. ScienceQA accuracy reaches 70.8%, marking a +2.6 pp improvement from the baseline (68.2%), confirming that ETS significantly benefits structured reasoning tasks as data availability expands. PubMedQA accuracy sees a notable rise to 93.2%, surpassing the base accuracy (91.9%) by +1.3 pp, further illustrating ETS's incremental performance gains in specialized domains.

Training with 100 Samples: When fine-tuned with 100 samples – a scenario still considered data-poor relative to typical fine-tuning paradigms – ETS consistently and markedly outperforms the base model. PIQA accuracy increases to 87.1%, a gain of +2.9 pp over the baseline (84.2%). OpenBookQA witnesses even larger improvements, reaching 84.9% accuracy, outperforming the baseline (81.3%) by +3.6 pp, highlighting ETS's capability to exploit limited yet increasing data resources effectively. RACE accuracy

achieves 86.2%, +2.6 pp above baseline performance (83.6%), clearly indicating ETS's consistent advantage in complex passage-level reasoning. ScienceQA exhibits substantial improvement (74.2%, +2.7 pp above baseline's 71.5%), further substantiating ETS's strengths in integrating incremental knowledge from additional training samples into structured scientific reasoning contexts. Lastly, PubMedQA results (94.3% vs. 93.0%) affirm ETS's steady benefit in expert-level biomedical reasoning, demonstrating consistent performance improvement of +1.3 pp even at relatively higher baseline accuracy levels.

These comprehensive low-resource experiments consistently demonstrate ETS's effectiveness in boosting performance beyond standard LLM fine-tuning across various domains and reasoning complexities. The incremental gains delivered by ETS become increasingly pronounced as the amount of available training data grows from extremely low (10 samples) to modestly low (100 samples). Across all benchmarks, ETS not only achieves higher accuracy rates but also demonstrates the framework's strong capacity to leverage even modest increases in available training data effectively. Crucially, these results underscore the robustness of ETS as a generalizable, low-resource methodology, highlighting its potential as a powerful approach for real-world applications where labeled data remains limited or expensive.

4.11. Human evaluation

To further examine the interpretability and usefulness of the generated rationales, we conducted a human evaluation experiment specifically focused on explanation quality. Two models were compared:

- **Base:** Standard fine-tuned LLM generating a single chain-of-thought rationale without structured elimination.
- **ETS (our):** Our full Eliminate-Then-Select framework, providing explicit elimination reasoning and calibrated explanations.

We randomly selected 50 questions from each of the six datasets (PIQA, OpenBookQA, RACE, ScienceQA, PubMedQA and EduExamQA), totaling 250 questions. Each question was answered by both models, and their generated explanations were anonymized and randomly ordered.

We recruited three human annotators who are graduate students familiar with educational testing and natural language processing. Annotators independently evaluated each explanation based on three criteria on a 5-point Likert scale (1 = poor, 5 = excellent):

- **Correctness:** Is the explanation factually correct?
- **Completeness:** Does the explanation cover why the chosen answer is correct and explicitly address incorrect alternatives?
- **Readability:** Is the explanation coherent, fluent, and easy to understand?

The final score for each criterion per question was averaged across annotators, with overall average scores computed per dataset and model. The results of the human evaluation are presented in Table 19. Across all datasets, ETS consistently achieves higher correctness scores ($\geq +0.42$ points average improvement per dataset). For instance, on PubMedQA, ETS explanations received a notably high correctness rating (4.68 vs. base 4.26), indicating significant reduction in factual inaccuracies. Annotators reported ETS's per-option reasoning explicitly addressed factual subtleties, reducing errors. Completeness is substantially enhanced by ETS ($\geq +0.52$ points improvement per dataset). The largest relative gain occurs on ScienceQA (ETS 4.35 vs. Base 3.43, +0.92 points). Annotators highlighted ETS's systematic elimination approach, consistently articulating why incorrect alternatives were invalidated, which significantly improves explanatory depth and educational value. ETS also delivers clear readability improvements ($\geq +0.35$ points increase per dataset). PIQA shows notable readability improvement (ETS 4.49 vs. Base 4.03, +0.46 points). ETS rationales, aided by calibration (Module D), often present coherent and polished final explanations, avoiding fragmented or unclear reasoning seen in the baseline. Averaging across all three criteria, ETS consistently outperforms the baseline by substantial margins ($\geq +0.45$ points overall). PubMedQA sees the highest overall improvement (ETS 4.61 vs. Base 4.10), underscoring ETS's effectiveness in domains demanding precise and reliable rationales.

This comprehensive human evaluation strongly validates ETS's superiority in generating high-quality, trustworthy explanations compared to standard LLM fine-tuning. ETS systematically addresses each answer choice, provides clearer rationale, and significantly reduces factual and logical errors. Such enhanced explanation quality directly improves ETS's educational utility, potentially fostering deeper student understanding and facilitating instructor trust. These human-rated results robustly confirm ETS's efficacy in producing high-quality explanations, complementing our quantitative analyses.

4.12. Inference cost & efficiency

Motivation: Module A in ETS performs per-option reasoning, which naturally raises the number of model calls relative to single-pass baselines. To make deployment trade-offs transparent, this section reports inference overhead – average calls per question, end-to-end latency, token usage, and dollar costs – measured under a unified protocol.

Metrics and instrumentation: For each question, we log:

- **Avg Calls/Q:** total API invocations across Modules A–D;
- **Latency (P50/P90):** wall-clock time from the first request to the last token received;
- **Prompt/Output Tokens:** token counts returned by the API;

Table 19
Human evaluation on all dataset.

Dataset	Method	Correctness	Completeness	Readability	Average
PIQA	Base	4.11	3.68	4.03	3.94
	ETS	4.57	4.48	4.49	4.51
OpenBookQA	Base	4.05	3.62	4.01	3.89
	ETS	4.52	4.44	4.43	4.46
RACE	Base	3.98	3.51	3.96	3.82
	ETS	4.49	4.40	4.38	4.42
ScienceQA	Base	3.92	3.43	3.89	3.75
	ETS	4.45	4.35	4.32	4.37
PubMedQA	Base	4.26	3.82	4.21	4.10
	ETS	4.68	4.60	4.56	4.61
EduExamQA	Base	4.02	3.58	3.98	3.86
	ETS	4.58	4.50	4.47	4.52

Table 20

Inference overhead vs. baselines (averaged over 6 datasets). Costs are derived from token counts using the representative input/output prices stated above.

Method	Avg Calls	Prompt Tok	Output Tok	Total Tok	Latency P50 (s)	Latency P90 (s)	Cost/1k Q (\$)
Direct MCQ	1.00	220	18	238	0.9	1.8	0.73
CoT-Once	1.00	240	120	360	1.5	3.1	1.80
Self-Consistency	10.00	2400	1200	3600	8.7	15.2	18.00
POSP	1.00	350	150	500	1.8	3.6	2.38
ETS-Lite (ours)	5.20	980	380	1360	3.2	6.1	6.25
ETS (ours)	7.20	1400	560	1960	4.6	8.9	9.10

- **Cost/1k Q:** computed from tokens via the model's list prices. For concreteness in this section we instantiate costs with representative values of 0.0025 per 1k input tokens and 0.010 per 1k output tokens; results scale linearly with the prices actually used in deployment.

Protocol and fairness: All systems use the same backbone model and decoding configuration (temperature/top-p/max tokens/stop sequences). Unless stated otherwise, we evaluate with batch size = 1, no client-side caching, and a single API region to control for network variability. Latencies are reported as median (P50) and tail (P90) across all questions, which better reflect user-perceived responsiveness than means. Token counts and costs are averaged per question.

Systems compared:

- **Direct MCQ:** single pass, no rationale.
- **CoT-Once:** one rationale for the predicted answer.
- **Self-Consistency (k = 10):** 10 sampled CoTs with majority vote.
- **Per-Option Single-Pass (POSP):** one call that evaluates all options in a single prompt.
- **ETS-Lite (ours):** a cost-lean ablation with one elimination round.
- **ETS (ours):** full ETS with threshold τ and at most R_{max} elimination rounds.

Reading Table 20. As expected, Self-Consistency is the most expensive approach: 10× calls inflate tokens and latency, leading to \$18.00 per 1k questions under our price instantiation. POSP is a strong single-call baseline with modest cost ($\approx$ \$2.38) but limited elimination control. ETS-Lite demonstrates that much of ETS's benefit is retained with one elimination round, roughly halving the overhead relative to full ETS (6.25 vs. 9.10), while preserving most of the reasoning structure. Full ETS incurs 7.2 calls per question on average – substantially cheaper than Self-Consistency – while offering fine-grained control over elimination and answer confirmation. In practice, these numbers translate into predictable budgeting: once token prices are fixed, cost scales nearly linearly with Avg Calls/Q.

To further capture cost-effectiveness, we compute cost-per-accuracy by dividing the cost per 1k questions by the average accuracy reported in main results. This metric highlights the trade-off between predictive benefit and monetary overhead. For example, while Self-Consistency achieves relatively strong accuracy, its cost-per-accuracy is the least favorable due to very high query costs. In contrast, ETS and ETS-Lite strike a better balance: their cost-per-accuracy ratios are substantially lower, showing that most of ETS's gains can be obtained at moderate additional cost relative to baselines.

Interpreting Table 21. Raising τ and allowing more rounds both increase calls and tokens, which in turn raises latency and cost. The growth is smooth, producing a usable cost-control surface: e.g., moving from ($\tau = 0.50$, $R_{max} = 1$) to (0.70, 2) adds $\approx$ +2.2 calls and $\approx$ +\$3.08 per 1k Q, a moderate increase for applications that benefit from additional scrutiny. Conversely, when budgets are tight, (0.60, 2) offers a balanced operating point: it delivers most of ETS's reasoning behavior while keeping overhead below \$8 per 1k Q.

Table 21

ETS operating points: overhead under different τ and R_{max} . We vary the elimination/answer threshold τ and the maximum number of elimination rounds R_{max} . Accuracy is reported in the main results table; here we isolate overhead.

τ (elimination/answer)	R_{max}	Avg Calls/Q	Latency P50 (s)	Total Tok/Q	Cost/1k Q (\$)
0.50	1	4.8	2.8	1260	5.70
0.60	2	6.1	3.7	1680	7.80
0.70	2	7.0	4.2	1890	8.78
0.75	3	7.8	4.9	2100	9.75

Table 22

Per-dataset patterns (overhead only).

Dataset	Method	Avg Calls/Q	Latency P50 (s)	Cost/1k Q (\$)
PIQA	POSP	1.0	1.5	2.00
	ETS-Lite	4.8	2.8	5.40
	ETS	6.6	4.1	8.10
RACE	POSP	1.0	2.3	2.70
	ETS-Lite	5.8	3.9	6.90
	ETS	8.4	5.4	10.20
ScienceQA	POSP	1.0	1.9	2.50
	ETS-Lite	5.4	3.3	6.50
	ETS	7.6	4.8	9.60
OpenBookQA	POSP	1.0	1.7	2.30
	ETS-Lite	5.1	3.0	6.00
	ETS	6.9	4.3	8.70
PubMedQA	POSP	1.0	1.8	2.40
	ETS-Lite	5.2	3.1	6.50
	ETS	7.1	4.5	9.00
EduExamQA	POSP	1.0	2.1	2.60
	ETS-Lite	5.6	3.5	6.80
	ETS	7.8	4.9	9.40

Table 23

Distribution of error sources.

Error source	% of Failures	Description
A. Reasoning error	43.0%	Incorrect or shallow CoT rationale leading to misjudgment per option.
B. Elimination threshold	18.5%	Incorrect option eliminated due to overly strict/lenient threshold.
C. Strategy misfire	14.0%	Over-pruning or under-pruning due to misjudged termination rounds.
D. Final decision error	24.5%	Correct option remained but not selected due to faulty confidence calibration or explanation inconsistency.

We observe consistent patterns across datasets of different difficulty and context length. On short-context commonsense sets (e.g., PIQA, OpenBookQA), POSP's single call already amortizes instruction overhead well, but ETS still offers structured elimination at a modest cost delta. On passage-heavy sets (e.g., RACE, ScienceQA), per-option reasoning steps tend to generate longer rationales and therefore more tokens; in these cases ETS-Lite can be an attractive compromise. The optional table below reports overhead statistics without repeating accuracy.

Observation (Table 22). Passage-heavy datasets (RACE/ScienceQA) unsurprisingly show higher latency tails and cost, because both per-option reasoning and final synthesis read longer contexts. This underscores the utility of configurable τ and R_{max} : they let practitioners cap overhead per question without redesigning prompts.

4.13. Error analysis

To understand the failure modes of the Eliminate-Then-Select (ETS) framework, we conduct a detailed error analysis across its four sequential modules: (A) per-option reasoning, (B) option elimination, (C) elimination strategy control, and (D) final selection and explanation generation. Since ETS is a pipeline-based approach, errors in early stages may propagate and affect downstream predictions. We analyze how and where such errors occur, quantify their contributions, and provide illustrative examples.

Error Categorization by Module: We randomly sample 200 failed predictions from the test splits across three datasets (RACE, ScienceQA, and OpenBookQA) and manually categorize the failure origin by module. Table 23 shows the distribution of error sources.

We observe that most failures originate from early stages – particularly Module A – highlighting the importance of accurate per-option reasoning. Later modules (C and D) exhibit fewer isolated errors but may amplify upstream issues.

Propagation and Cascading Effects: To analyze error propagation, we build a causal trace for each failed case: identifying the earliest erroneous decision and whether later modules could have recovered from it. Notably, among the 43% of cases with Module A failures, only 12% are “recoverable”—i.e., later modules corrected or bypassed the mistake.

Moreover, Module B errors tend to be non-recoverable — once the correct option is prematurely eliminated, the system cannot select it later regardless of strategy. In contrast, some Module D errors (e.g., due to explanation–answer mismatch) may be mitigated by better calibration.

Failure Case Examples: We include several representative cases. These examples illustrate different error modes:

- **Case 1 (Module A failure):** Misleading chain-of-thought reasoning caused the model to retain distractors that semantically contradicted the question.
- **Case 2 (Module B failure):** A correct option was eliminated due to an underestimated confidence score just below the threshold (e.g., 0.48 vs. $\tau = 0.5$).
- **Case 3 (Module C misfire):** The system exited early after two rounds while a better choice emerged in round 3.
- **Case 4 (Module D inconsistency):** The final answer was correct, but the explanation contradicted the rationale, leading to low faithfulness scores.

These findings suggest that improvements in per-option reasoning quality – e.g., using retrieval augmentation, rationale verification, or consistency training – could significantly improve overall performance. For Module B, we find that soft-voting or learned classifiers are more robust than fixed thresholds. For Modules C and D, adding confidence smoothing and self-consistency can reduce premature exit and improve alignment between explanation and answer.

4.14. Discussion and implications

This study introduces a novel Eliminate-Then-Select (ETS) framework for enhancing large language models (LLMs) in educational question answering, inspired by human test-taking behavior. Our findings have both theoretical and practical implications.

Theoretical Implications: From a theoretical perspective, this work contributes to the literature on explainable and reliable LLM-based reasoning in three key ways:

- **Human-Inspired Elimination Modeling:** While existing approaches often employ chain-of-thought (CoT) prompting, they typically focus on a single final answer and do not explicitly consider alternatives. Our ETS framework models the human process-of-elimination explicitly and modularly, which brings a new reasoning strategy into LLM architecture design.
- **Modular Reasoning Decomposition:** By decomposing reasoning into distinct modules (per-option reasoning, elimination filtering, dynamic control, and joint calibration), we present a flexible architecture that bridges symbolic process control with black-box neural reasoning. This advances the design of interpretable NLP systems.
- **Confidence Calibration Integration:** Unlike most prior QA frameworks, ETS integrates both local (per-option) and global (final answer) confidence estimation and calibration. This enables not just answer prediction, but also meta-cognitive awareness, which is crucial for building trustworthy educational AI.

Practical Implications: On the practical side, the ETS framework has direct utility for real-world applications:

- **Educational Technology:** In intelligent tutoring systems or exam prep tools, explainability is essential. ETS-generated rationales help learners understand why an answer is correct or incorrect, improving trust and learning outcomes.
- **Low-Resource Deployment:** Our results show that ETS performs robustly even in low-shot scenarios, reducing reliance on large-scale labeled data. This makes the approach suitable for niche educational domains (e.g., medical, legal) where annotated MCQs are scarce.
- **Model-Agnostic Design:** Since ETS treats the base LLM as a black box, it can be plugged into any state-of-the-art model without needing to retrain it. This makes it a lightweight, scalable solution that is compatible with commercial or open-source LLMs.

While prior methods such as standard CoT prompting or scoring-based elimination have demonstrated moderate improvements, they often conflate answer selection and explanation or require handcrafted rules for specific tasks. In contrast, our ETS framework introduces an explicit multi-stage process that evaluates each option, filters unlikely candidates, and dynamically controls strategy flow. This leads to more robust, interpretable, and calibrated performance, especially in challenging domains and ambiguous question scenarios.

4.15. Ethical considerations

As ETS is designed for educational applications, it is essential to address potential ethical implications. While the framework offers clear benefits in improving accuracy, explanation quality, and confidence calibration, improper use could lead to unintended consequences.

Risk of misuse: ETS could be misapplied as a shortcut for high-stakes examinations, such as standardized tests or medical licensing assessments. In such contexts, reliance on automated reasoning systems may undermine the integrity of the evaluation

process and distort the measurement of individual competence. Furthermore, unrestricted deployment risks enabling academic dishonesty, particularly if the system is accessible without safeguards.

Intended scope of use: The primary purpose of ETS is to support formative learning and educational research, where explanations foster deeper understanding and reflection. The framework is not designed as a replacement for human judgment in summative or credentialing examinations. Instead, its design emphasizes transparency and interpretability so that learners and educators can scrutinize the reasoning steps.

Safeguards and responsible deployment: To mitigate risks, deployment of ETS should incorporate:

- **Usage controls** — audit logs, access restrictions, and rate limits to prevent bulk exploitation in exam-like settings.
- **Transparency** — surfacing full rationales and evidence chains to encourage understanding rather than answer-copying.
- **Contextual restrictions** — prohibiting or discouraging deployment in high-stakes, licensure, or certification scenarios without human oversight.
- **Data responsibility** — handling only authorized and anonymized educational data, and adhering to privacy-preserving practices.

Broader impact: By emphasizing explanation quality and calibration, ETS may contribute positively to educational technology by promoting accountable AI and human-centric learning. However, ethical oversight, institutional approval, and human-in-the-loop supervision are prerequisites for its use in sensitive domains. We therefore frame ETS as a pedagogical assistant rather than an autonomous test-taking agent.

5. Conclusions

We presented Elimination before Selection, a novel framework that integrates large language models with human-like reasoning strategies to tackle multiple-choice questions. By decomposing the problem into modular steps – per-option reasoning, elimination filtering, strategy control, and answer/explanation calibration – our approach achieves both high accuracy and high interpretability. It leverages LLMs as powerful but black-box reasoners, steering them with structured prompts and lightweight learned components to emulate the process-of-elimination that students and test-takers use. The result is an LLM-based QA system that not only selects the right answer more often, but also shows its work: it tells you why each wrong option is wrong and how confident it is in the final choice.

Such a system can be transformative for educational technology. It can serve as an AI tutor that provides step-by-step explanations, helping learners understand the material rather than just giving answers. Teachers could use it to generate explanations for multiple-choice quizzes or to analyze common misconceptions (the rationales for eliminated options often highlight tempting wrong ideas that students might pick). The improved confidence calibration means the system knows when it might be guessing, which is crucial for safety — for example, in high-stakes domains like medicine, the system can refrain from giving a potentially harmful suggestion if it is not confident. Students can also learn effective problem-solving by example, seeing the elimination approach in action.

In conclusion, Elimination before Selection demonstrates a promising direction for aligning AI reasoning with human-like problem-solving strategies. It boosts performance on multiple-choice QA and, importantly, brings us closer to AI systems that can explain and educate. We hope this work encourages further exploration into interpretable reasoning processes for LLMs and their applications in education and beyond.

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CRediT authorship contribution statement

Mingsen Wang: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhaoqiang Li:** Resources, Project administration, Data curation. **Xuegang Zhao:** Visualization, Resources, Methodology, Data curation. **Qihan Guo:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link to my data in the paper.

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